

PROOF OF SERRANO'S CONJECTURE

ABSTRACT. We prove Serrano's conjecture on the ampleness of strictly nef adjoint divisors on smooth projective varieties. The result also holds for log canonical pairs.

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1. INTRODUCTION

A Cartier divisor is *strictly nef* if it has positive degree on every integral curve. This condition does not by itself imply ampleness. Serrano proposed that adding the canonical divisor changes this behavior in a precise uniform range; see [Ser95] and [CCP08, author final, Conjecture 0.1]. We prove the following result.

Theorem A. *Let X be a smooth integral projective variety of dimension $n \geq 1$, and let L be a strictly nef integral Cartier divisor on X . Then $K_X + tL$ is ample for every real number $t > n + 1$.*

The cone theorem gives strict nefness in this range. The remaining issue is bigness: a big adjoint in the range implies ampleness throughout it by basepoint-freeness and the positivity of L on every curve. We give the rational and real forms of this reduction in Proposition 2.3. For the study of strictly nef divisors and its log variants, we refer to [CCP08, LM23]; the original reduction and the abelian case appear in [Ser95, Lemma 1.3 and Proposition 1.4].

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1.1. Canonical varieties and the induction. We prove bigness by induction in a category that includes the canonical modifications of the subvarieties arising from jet kernels. The precise result is the following.

Theorem B. *Let X be a normal integral projective variety of dimension $n \geq 1$ with canonical singularities, and let L be a strictly nef integral Cartier divisor on X . Then $K_X + tL$ is ample for every real number $t > n + 1$. No $\mathbb{Q}$ -factoriality assumption is imposed.*

Theorem B includes every canonical variety needed in the induction, with no bound on its Cartier index. We prove the inductive assertion that $K_X + (2n + 1)L$ is big. The extremal-ray bound of [JLR26, Theorem 1.3] then gives the sharp nef endpoint $K_X + (n + 1)L$, and Lemma 2.5 recovers the stated ampleness range. The inductive assertion is eventual bigness for strictly nef Cartier divisors on all canonical varieties of smaller dimension. A generically finite morphism to a strictly nef variety then gives a fixed lower-dimensional adjoint polarization and a uniform bound for the degrees of covering curves; see Theorem 4.3. This permits the use of smooth resolutions of centers without assuming strict nefness on exceptional curves.

We now describe the contradiction used to prove this eventual bigness. If $D = K_X + (2n + 1)L$ is not big, choose an ample Cartier divisor A and normalize

$$(1.1) \quad P_j = r_j(D + u_j A), \quad r_j \rightarrow \infty, \quad u_j \rightarrow 0, \quad P_j^n \rightarrow \epsilon^n,$$

where $\epsilon > 0$ is fixed and small. The comparisons $P_j - r_j L$ nef and $P_j/r_j - K_X$ nef retain the canonical information, while $P_j \cdot C \geq r_j$ for every integral curve C . Lemma 2.6 gives the construction. Lower-dimensional canonical bigness excludes the bounded-degree centers that could obstruct single-point vanishing bounds or jets on the projective bundle over $X \times X$ defined in Set-up 7.1. The only remaining centers dominate both factors; branch and axis divisors reduce them to graphs of regular isomorphisms of finite covers, and an open orbit is excluded by Proposition 6.8.

Our argument adapts the moving-jet and Frobenius determinant constructions of [Ope26a, Ope26b]. We prove the local subadjunction, isolated-component, and full-kernel statements in Section 3. The determinant estimate in Theorem 9.2 allows negative cotangent slopes. The bound needed for those slopes follows from quantitative Harder–Narasimhan bend-and-break: on a resolving complete-intersection curve C of class ΛH^{n-1} , the allowed loss satisfies $\eta/(\Lambda H^n) = 2n/r_j$. The scalar jet estimate supplies one movable curve class that controls both determinant factors. After fixing the fiber coefficient, one index j , and all complex geometric data, we reduce modulo large primes. Full rank away from the Frobenius diagonal and a rank bound along it force incompatible vanishing orders of one determinant. Finally, Theorem 7.5 constructs the required two-slot jets with their fiber coefficient fixed before the normalization parameter tends to infinity. These statements are proved here with their indicated hypotheses; no numerical semiampleness or log abundance theorem is an additional input.

Pairs with real boundaries. The canonical result also gives the following consequence for klt pairs. For a real pair, we require the boundary to have finite support and the log canonical divisor to be $\mathbb{R}$ -Cartier.

Corollary 1.1. *Let (X, B) be a normal integral projective klt pair of dimension $n \geq 1$, where B is an effective real Weil divisor with finite support and $K_X + B$ is $\mathbb{R}$ -Cartier. Let L be a strictly nef $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor, and let $\ell \geq 1$ be any integer such that ℓL is an integral Cartier divisor. Then $K_X + B + tL$ is ample for every real $t > (n + 1)\ell$.*

Neither $\mathbb{Q}$ -factoriality nor separate $\mathbb{Q}$ -Cartierness of K_X is required. The factor ℓ records the integral lower bound for curve degrees of ℓL . The index of a rational log canonical divisor enters only when clearing denominators in a linear system.

For lc pairs, bigness must also be proved on every lc center. Quasi-log adjunction and the structure theorem for normal quasi-log canonical pairs give this additional assertion, leading to the following extension.

Corollary 1.2. *Let (X, B) be a normal integral projective log canonical pair of dimension $n \geq 1$, where B is an effective real Weil divisor with finite support and $K_X + B$ is $\mathbb{R}$ -Cartier. Let L be a strictly nef $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor, and let $\ell \geq 1$ be any integer such that ℓL is an integral Cartier divisor. Then $K_X + B + tL$ is ample for every real $t > (n + 1)\ell$.*

We prove Corollary 1.2 separately in Section 13, after the canonical induction. On the normalization of an lc center, the quasi-log structure gives a smooth model with nef moduli part and exceptional negative boundary. Keeping these terms permits bigness to descend, and the Reid–Fukuda basepoint-free theorem then applies to a nef and log big adjoint.

Structure of the paper. Section 2 proves the sharp recovery statement. Sections 3–5 develop the local tools, lower-dimensional estimate, and scalar jets. Sections 6 and 7 treat divisors, regular automorphisms of finite covers, and two-slot jets. Sections 8–10 prove the cotangent and determinant estimates and assemble the movable witness. Section 11 completes the canonical induction and proves Theorem A. Sections 12 and 13 prove the real-boundary consequences.

2. PRELIMINARIES

We adopt the standard notation and terminology for the minimal model program from [Sho92, KM98, BCHM10] and use them freely. We work over the field of complex numbers $\mathbb{C}$, except in the explicitly indicated reductions to positive characteristic.

An intersection of a Weil divisor with Cartier classes means the intersection of its cycle with those classes. We never use a pullback of a canonical Weil divisor unless it is $\mathbb{Q}$ -Cartier or the morphism is finite and the pullback is explicitly defined by divisorial valuations. Nefness, bigness, and ampleness of real divisors refer to their numerical classes. A sufficiently divisible multiple of a rational divisor always means a multiple for which the associated sheaf is invertible.

Notation 2.1. For a positive integer f , write CS_f for the following assertion: for every normal integral projective canonical f -fold T and every strictly nef integral Cartier divisor H on T , there exists a positive rational s such that $K_T + sH$ is big. We write $\text{CS}_{<n}$ for the collection of assertions CS_f with $1 \leq f < n$.

Theorem 2.2 (Cone and basepoint-free theorems). *For a smooth projective n -fold the negative extremal rays of the canonical divisor are generated by integral rational curves C with $0 < -K_X \cdot C \leq n + 1$. For an effective real lc pair (X, B) of dimension n , the corresponding bound is $0 < -(K_X + B) \cdot C \leq 2n$. In both cases the closed cone of effective curves is the closure of the sum of the nonnegative part and these rays.*

If (X, B) is a projective klt rational pair and J is a nef Cartier divisor such that $J - (K_X + B)$ is nef and big, then $|kJ|$ is basepoint-free for every sufficiently large integer k .

The cone bounds are used in the forms of [Ser95, Lemma 1.1] and [LM23, arXiv v2, Lemma 2.4]; the real lc cone decomposition is also given by [Fuj22, Theorem 7.7]. The last assertion is [Fuj12, Theorem 2.1], with its integer parameter equal to one; effectivity of the boundary makes its saturation condition automatic, as stated immediately before that result.

Proposition 2.3 (Sharp recovery). *Consider either of the following settings:*

- (1) X is smooth of dimension n , L is strictly nef Cartier, $A = K_X$, and $c = n + 1$;
- (2) (X, B) is a projective klt rational pair of dimension n , $A = K_X + B$ is $\mathbb{Q}$ -Cartier, L is a strictly nef $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -divisor, ℓL is integral Cartier, and $c = 2n\ell$.

Put $D_t = A + tL$. Then D_c is nef and D_t is strictly nef for every real $t > c$. The following conditions are equivalent: D_s is big for some real $s > c$; D_t is ample for every real $t > c$; the polynomial $(A + uL)^n$ is not identically zero; at least one number $A^j \cdot L^{n-j}$, $0 \leq j \leq n$, is nonzero. Thus failure of ampleness for even one $t > c$ forces all these mixed intersections to vanish and excludes bigness of every D_s with $s > c$.

Proof. We first locate the nef endpoint, then propagate bigness and recover ampleness. The rational Cartier index of A will enter only a multiplier.

Step 1. We prove nefness at c and strict nefness above c . In the smooth case, an integral curve has $L \cdot C \geq 1$. In the rational-pair case it has $L \cdot C \geq 1/\ell$. Theorem 2.2 therefore makes $A + cL$ nonnegative on each negative extremal generator, and it is nonnegative on the A -nonnegative part since L is nef. Continuity gives D_c nef. For $t > c$,

$$(2.1) \quad D_t = D_c + (t - c)L$$

is nef and has positive degree on every integral curve.

Step 2. We propagate one big adjoint to every parameter above c . Assume D_s is big for some $s > c$. For $c < r < s$, set $\alpha = (r - c)/(s - c)$. Then $D_r = \alpha D_s + (1 - \alpha)D_c$ is big. A positive multiple of an interior point of the pseudo-effective cone plus a point of that cone remains in its interior: translate an open neighborhood contained in the cone and use closure under addition. For $r \geq s$, use $D_r = D_s + (r - s)L$. Thus D_r is nef and big for every real $r > c$.

Step 3. We prove ampleness for rational parameters and then for real parameters. Write $r = a/b > c$ with positive integers a, b , and choose $h \geq 1$ such that hA is Cartier. Take an integer $m \geq 2$ divisible by h and $b\ell$, with $h = \ell = 1$ in the smooth case. Then

$$J = mD_r = mA + \frac{ma}{b\ell}(\ell L), \quad J - A = (m - 1)D_r + rL$$

satisfy: J is integral Cartier and nef, and $J - A$ is a nef and big rational Cartier divisor. Theorem 2.2 makes $|kJ|$ basepoint-free for large k . Its morphism to its projective image has no positive-dimensional fiber, because such a fiber contains an integral curve of J -degree zero. The morphism is proper and quasi-finite, hence finite. The finite pullback of $\mathcal{O}(1)$ is ample, so D_r is ample.

For a real $t > c$, choose rationals $c < r_- < t < r_+$. With $\beta = (r_+ - t)/(r_+ - r_-)$ we have $D_t = \beta D_{r_-} + (1 - \beta)D_{r_+}$, which is ample by convexity. Conversely, an ample divisor is big.

Step 4. We identify the numerical obstruction. Multilinearity gives

$$(2.2) \quad (A + uL)^n = \sum_{j=0}^n \binom{n}{j} u^{n-j} (A^j \cdot L^{n-j}).$$

The polynomial vanishes identically exactly when all its coefficients vanish. If it is nonzero, choose a rational $r > c$ outside its finitely many roots. Since D_r is nef, $D_r^n \geq 0$, and the choice gives $D_r^n > 0$. The nef top-intersection criterion makes D_r big. Conversely an ample rational D_r has positive top intersection. This proves all equivalences and their asserted contrapositive. $\square$

Theorem 2.4 ([JLR26, Theorem 1.3]). *Let (X, Δ) be an lc pair over an algebraically closed field of characteristic zero. Every $(K_X + \Delta)$ -negative extremal ray of the closed cone of effective curves has a rational generator C satisfying $-(K_X + \Delta) \cdot C \leq \dim X + 1$.*

Lemma 2.5 (Sharp recovery for pairs). *Let (X, B) be a projective lc real pair of dimension $n \geq 1$, and let H be strictly nef integral Cartier. Put $R = K_X + B$ and $c = n + 1$. Then $R + cH$ is nef. If $R + TH$ is ample for one real $T > c$, then $R + sH$ is ample for every real $s > c$. If the pair is klt and rational, it suffices that $R + TH$ be big for one real $T > c$.*

Proof. By Theorem 2.4, each negative ray has a rational generator with $-R \cdot C \leq c$, whereas $H \cdot C \geq 1$. The cone decomposition therefore makes $N = R + cH$ nef. For $c < s < T$,

$$(2.3) \quad R + sH = \frac{s - c}{T - c}(R + TH) + \frac{T - s}{T - c}N.$$

A positive multiple of an ample class plus a nef class is ample. For $s \geq T$, add $(s - T)H$ to $R + TH$. This argument applies directly to real boundaries.

In the rational klt case, (2.3) first propagates bigness from T to all $s > c$, and $R + sH = N + (s - c)H$ is strictly nef. For rational $s > c$, choose an integer $m \geq 2$ clearing the denominators of $R + sH$. The nef Cartier divisor $J = m(R + sH)$ satisfies $J - R = (m - 1)(R + sH) + sH$ nef and big. Theorem 2.2 gives a basepoint-free multiple. Its morphism is finite because J is strictly nef, and hence $R + sH$ is ample. Positive interpolation between rational parameters proves the real conclusion. $\square$

Lemma 2.6 (Normalization of a nonbig adjoint). *Let X be canonical of dimension $n \geq 2$, let L be strictly nef integral Cartier, and assume $D = K_X + (2n + 1)L$ is not big. Given an ample Cartier divisor A and a real $\epsilon > 0$, there exist positive rational $u_j \rightarrow 0$ and positive integers $r_j \rightarrow \infty$ such that the ample rational divisors $P_j = r_j(D + u_j A)$ satisfy*

$$(2.4) \quad P_j^n \rightarrow \epsilon^n, \quad P_j - r_j L \text{ is nef}, \quad P_j/r_j - K_X \text{ is nef}.$$

For every integral curve $C \subset X$, $P_j \cdot C \geq r_j$.

Proof. Proposition 2.3 gives $N = K_X + 2nL$ nef and $D = N + L$ nef, so $D^n = 0$. Choose positive rational u_j decreasing to zero, and put $w_j = (D + u_j A)^n$. Then $w_j > 0$ and $w_j \rightarrow 0$. Take r_j to be the nearest positive integer to $\epsilon/w_j^{1/n}$, deleting finitely many terms if necessary. Thus $r_j w_j^{1/n} \rightarrow \epsilon$ and $r_j \rightarrow \infty$. The comparisons are

$$P_j - r_j L = r_j(N + u_j A), \quad P_j/r_j - K_X = (2n + 1)L + u_j A.$$

They are nef, and the first gives $P_j \cdot C \geq r_j L \cdot C \geq r_j$. $\square$

3. SUBADJUNCTION AND FULL MOVING KERNELS

In this section, we prove the local results used to bound centers arising from complete jet kernels, including their restrictions to fibers. The arguments develop the local constructions of [Ope26a, Lemmas 7.1–7.4], also used in [Ope26b, Lemmas 4.3–4.6], with the generic lc and ordinary-ideal hypotheses made explicit.

3.1. Numerical subadjunction at a generic lc center. We first prove a numerical subadjunction statement with log canonicity required only near the generic point of the center. We use the following precise forms of dlt modification, adjunction, and the canonical bundle formula.

Theorem 3.1 ([FH23, Theorem 2.10 and Remark 2.11]). *Let Z be normal quasi-projective and $\Theta \geq 0$ an $\mathbb{R}$ -divisor with $K_Z + \Theta$ $\mathbb{R}$ -Cartier. There exists a projective birational $p: Q \rightarrow Z$ with Q $\mathbb{Q}$ -factorial such that every exceptional prime has discrepancy at most -1 . If $K_Q + \Gamma = p^*(K_Z + \Theta)$, then $\Gamma \geq 0$, the coefficientwise truncation $B_Q = \min\{\Gamma, 1\}$ defines a dlt pair, and $E_Q = \Gamma - B_Q \geq 0$. The coefficients can be taken rational for rational input.*

Theorem 3.2 ([Fuj07, Propositions 3.9.2 and 3.9.4]). *The lc centers of an effective dlt pair are its strata obtained as irreducible components of intersections of coefficient-one components. Every stratum S is normal. Iterated adjunction gives an effective rational different Δ_S satisfying $(K_Q + B_Q)|_S \sim_{\mathbb{Q}} K_S + \Delta_S$ for rational input. The pair (S, Δ_S) is dlt and its lc centers are the proper strata contained in S .*

Theorem 3.3 ([FG14, Definition 3.1 and Theorem 3.5]; [Amb04, Theorem 2.7]). *Let $q: (S, B_S) \rightarrow W$ be a proper surjective morphism of normal varieties with connected fibers, where $K_S + B_S$ is $\mathbb{Q}$ -Cartier. Assume the pair is sub-klt over the generic point of W , the discrepancy b -divisor $\mathbf{A}(S, B_S)$ satisfies*

$$(3.1) \quad \text{rank } q_* \mathcal{O}_S([\mathbf{A}(S, B_S)]) = 1,$$

and $K_S + B_S \sim_{\mathbb{Q}} q^* D'$ for a rational Cartier class D' on W . Then

$$(3.2) \quad D' \sim_{\mathbb{Q}} K_W + B_W + M_W,$$

where the coefficient of a prime F in B_W is $1 - t_F$, with t_F the sub-lc threshold of q^*F over its generic point. The moduli b -divisor descends to a nef rational Cartier divisor on a higher proper birational model when W is complete. Global sub-lc singularities are not required.

Here the sheaf in (3.1) is computed by the discrepancy valuation inequalities, or on a log resolution and then pushed down; this fixes the convention in [FG14, Sections 2.1–2.4].

Proposition 3.4 (Numerical generic subadjunction). *Let Z be normal projective $\mathbb{Q}$ -factorial and klt, let $\Theta \geq 0$ be rational, and let $V \subset Z$ be integral of dimension $f > 0$. Assume (Z, Θ) is lc on an open neighborhood of the generic point of V and has an lc place with center V . Let P be a nef rational Cartier divisor on V , and let $\rho: V^* \rightarrow V$ be any smooth projective resolution through the normalization. Then*

$$(3.3) \quad K_{V^*} \cdot (\rho^*P)^{f-1} \leq (K_Z + \Theta)|_V \cdot P^{f-1}.$$

No $\mathbb{Q}$ -Gorenstein assumption on the normalization of V is imposed.

Proof. Put $R = K_Z + \Theta$. We retain the non-lc contribution on a dlt modification and use the finite Stein base of a minimal stratum. All subsequent canonical intersections on normal bases are intersections of Weil cycles.

Step 1. We construct an effective pair on a stratum dominating V . Theorem 3.1 gives

$$(3.4) \quad K_Q + \Gamma = p^*R, \quad B_Q = \min\{\Gamma, 1\}, \quad E_Q = \Gamma - B_Q \geq 0.$$

Every strict component of Γ has the nonnegative coefficient of Θ , and every exceptional component has coefficient at least one. Choose an open U containing the generic point of V where the pair is lc. A component of E_Q has discrepancy less than -1 , so its image lies in $\text{Nlc}(Z, \Theta)$ and $E_Q|_{p^{-1}(U)} = 0$.

Properness gives a center on Q for the given lc place. Over U it is an lc center of the crepant dlt pair. Theorem 3.2 and closure of the corresponding floor intersection give a global stratum mapping onto V . Among these finitely many strata choose S minimal by inclusion. It is normal, integral, and projective. Write $g = p|_S$. Adjunction gives $K_S + \Delta_S \sim_{\mathbb{Q}} (K_Q + B_Q)|_S$ with $\Delta_S \geq 0$ and dlt. The stratum S meets $p^{-1}(U)$, so it is not contained in E_Q . Since Q is $\mathbb{Q}$ -factorial, a Cartier multiple of E_Q restricts to an effective Cartier divisor on S . Thus

$$(3.5) \quad B_S = \Delta_S + E_Q|_S \geq 0, \quad K_S + B_S \sim_{\mathbb{Q}} g^*(R|_V).$$

These are rational line-bundle identities even when a divisor representative contains V .

Step 2. We obtain generic klt singularities on the correct connected-fiber base. Every lc center of (S, Δ_S) is a proper stratum inside S . None dominates V by minimality. Removing their finitely many proper closed images, together with the image of the non-lc surplus, makes (S, B_S) klt over a nonempty open of V . Factor through normalization and take Stein factorization:

$$(3.6) \quad S \xrightarrow{q} W \xrightarrow{\zeta} V^\nu \xrightarrow{\nu} V.$$

Then W is normal projective, ζ is finite of degree $e > 0$, and $q_*\mathcal{O}_S = \mathcal{O}_W$. Put $D_V = \nu^*(R|_V)$ and $D' = \zeta^*D_V$. By (3.5), $K_S + B_S \sim_{\mathbb{Q}} q^*D'$, and the pair is klt over the generic point of W .

Step 3. We verify the rank-one hypothesis of the canonical bundle formula. On an open W° where $(S^\circ, B_S|_{S^\circ})$ is klt, its boundary is effective. Every discrepancy a_F is greater than -1 , so $\lceil a_F \rceil \geq 0$; on a nonexceptional divisor it is the round-up of a number in $(-1, 0]$, hence zero. The discrepancy sheaf on S° is therefore $\mathcal{O}_{S^\circ}$. A rational function satisfying its inequalities has no pole in codimension one and is regular by normality, while every regular function satisfies the nonnegative exceptional inequalities. Equivalently, on a log resolution all permitted poles are exceptional and disappear on pushforward. Consequently the direct image of this sheaf restricts to $q_*\mathcal{O}_{S^\circ} = \mathcal{O}_{W^\circ}$ and has generic rank one. Theorem 3.3 gives (3.2) on W .

Step 4. We prove effectivity of the discriminant on the original Stein base. Fix a prime $F \subset W$ and work near its generic point, where it is Cartier. Some prime $T \subset S$ dominates F

with positive integral multiplicity m in q^*F : the zero locus of the pulled-back local equation is Cartier and its finitely many proper images cover F . If b is the coefficient of T in B_S , then $b \geq 0$. Sub-lc singularities of $B_S + tq^*F$ require $b + tm \leq 1$, whence $t \leq (1 - b)/m \leq 1$. Thus $t_F \leq 1$ and $B_W \geq 0$.

The threshold is finite and rational even when the special fiber is not lc. On a log resolution of $B_S + q^*F$ near the generic point of F , horizontal coefficients are less than one, and each vertical divisor has positive multiplicity in the pullback of F . Sufficiently negative t makes all finitely many vertical coefficients at most one. The threshold is the minimum of their rational upper bounds. Negative thresholds therefore cause no difficulty. This proves effectivity on W ; discriminant traces on higher models may have negative exceptional coefficients.

Step 5. We test the moduli term on a model where it is nef. Write $P_\nu = \nu^*P$ and $P_W = \zeta^*P_\nu$. Choose a smooth projective model $\mu: H \rightarrow W$ dominating a model where the moduli b-divisor descends, using Chow's lemma [Sta26, Lemma 30.18.1, Tag 0200] and resolution if necessary. Its trace M_H is nef and rational Cartier, and $\mu_*M_H = M_W$ as Weil cycles. Hence

$$(3.7) \quad \begin{aligned} M_W \cdot P_W^{f-1} &= M_H \cdot (\mu^*P_W)^{f-1} \geq 0, \\ B_W \cdot P_W^{f-1} &\geq 0, \quad D' \cdot P_W^{f-1} \geq K_W \cdot P_W^{f-1}. \end{aligned}$$

The first inequality follows by ample approximation of nef classes; the second follows on each effective prime cycle. No pullback or nefness of M_W on W is asserted. In particular, negative exceptional discriminants on H have not been discarded under an effectivity assertion.

Step 6. We descend the canonical cycle through the finite map. Choose canonical divisors from a rational top form on V^ν and its pullback. At divisorial discrete valuation rings the finite separable map ζ gives

$$K_W = \zeta_{\text{Weil}}^* K_{V^\nu} + R_\zeta, \quad R_\zeta \geq 0, \quad \zeta_* K_W = e K_{V^\nu} + \zeta_* R_\zeta.$$

The valuation pullback is defined without $\mathbb{Q}$ -Cartierness of K_{V^ν} ; a ramification index a contributes $a - 1$ to the differential. The degree formula for valuations gives the pushforward identity. Projection formula in (3.7) implies

$$(3.8) \quad e D_V \cdot P_\nu^{f-1} \geq e K_{V^\nu} \cdot P_\nu^{f-1} + (\zeta_* R_\zeta) \cdot P_\nu^{f-1} \geq e K_{V^\nu} \cdot P_\nu^{f-1}.$$

Step 7. We compare with the prescribed smooth resolution. Write $\rho = \nu \circ a$. For compatible canonical divisors, $a_* K_{V^*} = K_{V^\nu}$, because a is an isomorphism at generic divisorial points and exceptional prime cycles push to zero. Thus

$$K_{V^*} \cdot (a^* P_\nu)^{f-1} = K_{V^\nu} \cdot P_\nu^{f-1}, \quad D_V \cdot P_\nu^{f-1} = (R|_V) \cdot P^{f-1}.$$

For $f = 1$, the birational map between normal curves is an isomorphism and these are degree equalities. Dividing (3.8) by e proves (3.3). The argument retains the non-lc surplus in B_S , verifies generic klt and rank one, and uses only canonical Weil-cycle pushforwards on the center. $\square$

3.2. Isolated base components. We now derive the degree and canonical bounds from the ordinary multiplicity of an actual linear subsystem.

Proposition 3.5. *Let Z be normal projective $\mathbb{Q}$ -factorial klt of dimension $d \geq 1$, let R be a nef rational Cartier divisor, and let $0 \neq H \subset H^0(Z, \mathcal{O}_Z(kR))$, where kR is integral Cartier. Let $\mathfrak{b}$ be its base ideal and assume the base locus is proper. Let V be an integral component of dimension $0 \leq f < d$, with smooth generic point on Z , such that $\mathfrak{b}\mathcal{O}_{Z, \eta_V}$ is primary for the maximal ideal $\mathfrak{m}$. Put $a = d - f$. If $\mathfrak{b}\mathcal{O}_{Z, \eta_V} \subset \mathfrak{m}^h$ for an integer $h \geq 1$, then*

$$(3.9) \quad R^f \cdot V \leq (k/h)^a R^d.$$

For $f = 0$, the left side is one. There exist $\lambda \in \mathbb{Q}_{>0}$ with $\lambda \leq a/h$, an integer $N > \lambda$, and actual members $D_1, \dots, D_N$ of $\mathbb{P}(H)$ such that

$$(3.10) \quad \Theta = \frac{\lambda}{N} \sum_{j=1}^N D_j \geq 0, \quad \Theta \sim_{\mathbb{Q}} cR, \quad c = k\lambda \leq ak/h,$$

and (Z, Θ) is lc near η_V with an lc place centered on V . If $f > 0$, every smooth resolution $\rho: V^* \rightarrow V$ through normalization satisfies

$$(3.11) \quad K_{V^*} \cdot (\rho^*(R|_V))^{f-1} \leq (K_Z + cR)|_V \cdot (R|_V)^{f-1}.$$

Neither completeness of H nor ampleness, bigness, or semiampleness of R is required.

Proof. We prove (3.9) by local length and global intersections. We then realize the generic ideal threshold by averaging members of H and use Proposition 3.4 for (3.11).

Step 1. We choose parameter sections in the given subsystem. Let $A = \mathcal{O}_{Z, \eta_V}$, a regular local ring of dimension a , with residue field $\mathbb{C}(V)$. Trivialize kR there. The local representatives of H generate an $\mathfrak{m}$ -primary ideal $I \subset \mathfrak{m}^h$. Choose $g_1, \dots, g_a$ as local representatives of sections of H inductively. After $i - 1$ choices, the quotient by the regular sequence is Cohen–Macaulay of positive dimension. Its finitely many associated primes are different from $\mathfrak{m}$ and none contains I . For each such prime the sections of H lying in it form a proper complex linear subspace. Avoiding their union chooses g_i as a nonzerodivisor. After a choices the ideal $\mathfrak{q} = (g_1, \dots, g_a)$ is primary and the sequence is regular. This is the Cohen–Macaulay parameter criterion of [Sta26, Lemma 10.104.2, Tag 02JN]. Additional finitely many proper linear conditions needed for the global cuts can be avoided simultaneously.

Step 2. We prove the local multiplicity bound h^a . Write $\ell_{\mathfrak{q}} = \text{length}_A(A/\mathfrak{q})$. The regular-sequence associated graded identity [Sta26, Lemma 10.69.2, Tag 00LN] and the corresponding identity for the regular local maximal ideal give

$$\begin{aligned} \text{gr}_{\mathfrak{q}} A &= (A/\mathfrak{q})[T_1, \dots, T_a], & \text{length}(A/\mathfrak{q}^j) &= \ell_{\mathfrak{q}} \binom{j+a-1}{a}, \\ \text{gr}_{\mathfrak{m}} A &= \mathbb{C}(V)[U_1, \dots, U_a], & \text{length}(A/\mathfrak{m}^{hj}) &= \binom{hj+a-1}{a}. \end{aligned}$$

Since $\mathfrak{q}^j \subset \mathfrak{m}^{hj}$, comparison after division by j^a and passage to the limit gives $\ell_{\mathfrak{q}} \geq h^a$. The local Cartier complete-intersection multiplicity is this length, by [Sta26, Lemmas 43.16.1 and 43.16.5, Tags 0B02 and 0B06].

Step 3. We control the global cuts without losing the component V . Put $B = \text{Supp}(\mathcal{O}_Z/\mathfrak{b})$ and start with $T_0 = [Z]$. At stage i , delete from the effective cycle T_{i-1} exactly the components contained in B , obtaining U_{i-1} . Choose the divisor D_i of the section for g_i to meet each component of U_{i-1} properly; every corresponding avoidance condition is proper because that component is not in the base locus. Put $T_i = D_i \cdot U_{i-1}$.

No deleted component contains η_V . Its dimension before stage $i \leq a$ is $d - i + 1 \geq f + 1$, so such containment would give a base component properly containing V . Thus the local complete intersection at η_V is retained, and the coefficient of V in T_a is $\ell_{\mathfrak{q}} \geq h^a$. Do not delete the zero-dimensional components after the final cut when $f = 0$. Nefness gives

$$R^{d-i} \cdot T_i = kR^{d-i+1} \cdot U_{i-1} \leq kR^{d-i+1} \cdot T_{i-1},$$

and hence

$$h^a R^f \cdot V \leq \ell_{\mathfrak{q}} R^f \cdot V \leq R^f \cdot T_a \leq k^a R^d.$$

Nonnegativity on all effective cycles follows by ample approximation of R . This proves (3.9), including the ordinary zero-cycle degree when $f = 0$.

Step 4. We compute the threshold at the generic point of V . Shrink to a smooth neighborhood U of η_V with $V \cap U$ smooth and base support exactly $V \cap U$. Take a log resolution $\mu: W \rightarrow Z$ of the ideal and the variety, with

$$\mathfrak{b}\mathcal{O}_W = \mathcal{O}_W(-F), \quad F \geq 0,$$

and simple normal crossing support including the exceptional and relative-canonical divisors. Factoring the pulled-back evaluation by F makes the moving subsystem generate $\mathcal{O}_W(\mu^*(kR) - F)$ everywhere. This concerns H itself.

If $r_j = \text{ord}_{E_j}(\mathfrak{b})$ and A_j is the log discrepancy, the resolution formula [Mus12, 2011 author manuscript, Theorem 1.1] gives

$$(3.12) \quad \lambda = \min_{\substack{\eta_V \in \mu(E_j) \\ r_j > 0}} \frac{A_j}{r_j} > 0.$$

Only finitely many resolution divisors occur. Each has positive discrepancy, and a minimizing divisor has center V , since its center both contains η_V and lies in the local base support. If $a \geq 2$, blowing up the smooth center gives discrepancy a and ideal order at least h , so $\lambda \leq a/h$. For $a = 1$, the valuation of V itself has discrepancy one and order at least h ; no exceptional divisor from blowing up a Cartier center is asserted.

Step 5. We realize the threshold by an average of actual divisors. Choose an integer $N > \lambda$ and N general members M_j of the generated moving subsystem on W . Successive Bertini choices along the finitely many strata make their union with the fixed support simple normal crossing over a smaller U , with reduced moving members, no repeated component, and no fixed component contained in a moving member. If the subsystem has dimension zero, its generated line bundle has its unique member equal to zero, which satisfies the requirement. The members come from actual sections of H and

$$\mu^*D_j = F + M_j.$$

Define Θ and c by (3.10). For the pullback pair $K_W + B_W = \mu^*(K_Z + \Theta)$, a fixed component has coefficient

$$1 - A_j + \lambda r_j \leq 1,$$

where irrelevant centers can be removed by shrinking U . When $r_j = 0$, its coefficient is $1 - A_j < 1$. Every moving coefficient is $\lambda/N < 1$. The simple normal crossing subpair is lc precisely under these upper coefficient bounds, even if some exceptional coefficients are negative. At a minimizing divisor in (3.12) the coefficient is exactly one. This proves the stated local lc property and lc place. The resolution-and-Bertini mechanism is also described in [Mus12, 2011 author manuscript, Example 1.10]; averaging allows λ to exceed one.

Step 6. We obtain the canonical inequality by generic subadjunction. For $f > 0$, Proposition 3.4 applies to (Z, Θ) , V , and $R|_V$, because all its local lc and lc-place hypotheses were proved in Step 5. It gives (3.11) using $\Theta \sim_{\mathbb{Q}} cR$. Thus the generic-subadjunction premise needed for the canonical consequence is discharged by Proposition 3.4; the degree and divisor-realization arguments are independent of it. $\square$

3.3. The entire kernel and restriction to a fiber. We prove the parameter-differentiation statement for the entire kernel, following the method of [EKL95, Lemma 2.1 and Proposition 2.3]. The ordinary ideal and restriction assertions will be used explicitly.

Proposition 3.6 (Full moving kernels). *Let Z be integral projective of dimension d , let P be nef, big, and semiample rational Cartier, and let kP be Cartier. Let $\tau_0, \dots, \tau_d > 0$ have constant gap $\delta > 0$, with $k\delta \geq 4$. For smooth x put*

$$(3.13) \quad H_i(x) = H^0(Z, \mathcal{O}_Z(kP) \otimes \mathfrak{m}_x^{q_i}), \quad q_i = \lceil k\tau_i \rceil, \quad h = \lceil k\delta/2 \rceil.$$

Assume all $H_i(x)$ are nonzero at very general x and the lowest base support has a positive-dimensional component through x . Then $d \geq 2$. After shrinking, there exist a nonempty open $U \subset Z_{\text{reg}}$, a smooth integral T with finite etale surjective mark map $x: T \rightarrow U$, an adjacent pair $i, i+1$, and an integral closed incidence $\Gamma \subset T \times Z$ with the following properties.

- (1) $\Gamma \rightarrow T$ is flat and projective with geometrically integral fibers V_t of dimension $1 \leq f < d$. Each V_t contains $x(t)$ and is a component of both base supports. The evaluation $\Gamma \rightarrow Z$ is dominant.
- (2) The higher kernels are the fibers of a vector subbundle $\mathcal{K} \subset H^0(Z, kP) \otimes \mathcal{O}_T$. Its full evaluation ideal $\mathfrak{b}_T$ satisfies $\mathfrak{b}_T \subset I_\Gamma^h$ in ambient neighborhoods of a dense open $\Omega \subset \Gamma \cap (T \times Z_{\text{reg}})$. At those points the fiber base support is exactly V_t , and its full ideal satisfies $\mathfrak{b}_t \subset I_{V_t}^h$. At η_{V_t} it is maximal-ideal primary. Moreover $k/h \leq 2/\delta$.
- (3) Let $g: Z \rightarrow B$ be fixed. Choose $(t, z) \in \Omega$ where $V_t \rightarrow W_t = g(V_t)_{\text{red}}$ is smooth and $b = g(z)$ is smooth on W_t . Let F be the reduced fiber component through z . Near z ,

$$(3.14) \quad \mathfrak{b}_t \mathcal{O}_{Z_b} \subset I_{F/Z_b}^h,$$

with support exactly F . If C is an integral reduced component of Z_b containing F properly, the image series

$$(3.15) \quad H_C = \text{im} (H_{i+1}(x(t)) \rightarrow H^0(C, \mathcal{O}_Z(kP)|_C))$$

is nonzero and has F as a generically isolated base component with ideal contained in $I_{F/C}^h$. Such admissible points are dense after further shrinking. The image series need not be an intrinsic full jet kernel on C .

- (4) If a semiample morphism $\phi: Z \rightarrow Y$ with $mP = \phi^*A$, A ample Cartier, is an isomorphism on an open meeting V_t , then $P^f \cdot V_t > 0$.

The kernel, ordinary-ideal, and restriction assertions use only that $\mathcal{O}_Z(kP)$ is a line bundle. Numerical subadjunction requires separate checks of the singularities and projectivity of the ambient varieties involved.

Proof. Write $E = \mathcal{O}_Z(kP)$ and $W = H^0(Z, E)$. We construct the algebraic family before differentiating its full kernel frame.

Step 1. We spread an adjacent common component of the base schemes. Over the smooth marked-point open, the jet bundle of length q_i is obtained from the $(q_i - 1)$ st infinitesimal neighborhood of the graph of the mark. Its local basis consists of monomials of total degree less than q_i in d transverse regular parameters, so that neighborhood is finite flat over the mark. Remove the finitely many rank-jump loci of the maps from $W \otimes \mathcal{O}$ to these jet bundles. Their kernels $\mathcal{K}_i$ are vector subbundles, their formation commutes with base change, and their fibers are exactly $H_i(x)$. Each has positive rank by the very-general hypothesis.

Evaluation gives projective relative base schemes $\mathcal{B}_i \subset U \times Z$, with nested supports on fibers. All contain the mark, and all are proper since they have a nonzero defining section on the integral Z . The positive local dimension of $\mathcal{B}_0(x)$ at the mark holds on an open after shrinking: the condition is constructible by fiber-dimension semicontinuity along the marked section and contains very general points.

At the geometric generic mark choose a positive-dimensional component V_0 through it and successively components V_j of the higher base supports containing V_{j-1} . Their dimensions lie in $\{1, \dots, d-1\}$. Among $d+1$ such components an adjacent inclusion has equal dimension, hence is equality. The components and their equations, including the other generic components of the two relevant supports, are defined over a finite extension of $\mathbb{C}(U)$. Normalize U in this extension and shrink to make the resulting $T \rightarrow U$ finite etale. Let Γ be the reduced closure of the chosen component in $T \times Z$. It contains the marked section by generic containment and closure.

Close every generic component of the two base supports and delete vertical components by shrinking T . Generic flatness and openness of geometric integrality make the reduced closures flat with integral fibers of fixed dimensions. The intersections of distinct closures have generically

smaller dimension whenever containment could occur; upper semicontinuity for these projective intersections excludes new containments on fibers after another shrink. Thus the selected fibers really are components of both supports. Evaluation is dominant since Γ contains the moving marked section. Delete its intersections with all other higher-base components. The local relative support is then Γ alone and the fiber support is V_t alone. At η_{V_t} the radical is the maximal ideal; the quotient is a zero-dimensional Noetherian local ring, hence Artinian, so the localized ideal is primary.

Step 2. We differentiate every higher-kernel section into the lower kernel. On a smooth affine parameter open choose a frame of $\mathcal{K}_{i+1}$. A frame section is

$$s(t, z) = \sum_{\alpha} c_{\alpha}(t) s_{\alpha}(z)$$

in a fixed basis of W . A parameter derivation differentiates only the c_{α} , so every iterated derivative remains a family of global sections of E . This is independent of the trivialization of E , whose transitions depend only on z .

If Δ_x is the graph of the mark, the kernel property gives $s \in I_{\Delta_x}^{q_{i+1}} E$. The Leibniz rule gives $D(I^v) \subset I^{v-1}$ for every ideal. Thus a derivative of order $r \leq q_{i+1} - q_i$ specializes into the full lower kernel and vanishes on V_t , hence on the reduced incidence. The rounding checks are

$$q_{i+1} - q_i \geq k\delta - 1, \quad h - 1 < k\delta/2 \leq k\delta - 1.$$

Every derivative of order less than h therefore vanishes along Γ , and $k/h \leq 2/\delta$. This applies to every member of the frame, not to a selected divisor.

Step 3. We turn parameter derivatives into ordinary ideal powers. Restrict to the smooth locus of Γ in $T \times Z_{\text{reg}}$ where the dominant evaluation to Z is smooth. Its tangent map gives

$$T_{(t,z)}(T \times Z) = T_{(t,z)}\Gamma + (T_t T \times \{0\}).$$

Thus parameter directions span the normal space. Choose $a = d - f$ parameter derivations whose normal images are independent and shrink where their determinant is nonzero. At the generic point of Γ , the ring $R = \mathcal{O}_{T \times Z, \eta_{\Gamma}}$ is regular of dimension a , with maximal ideal $\mathfrak{m} = I_{\Gamma} R$ and residue field K . We have $\text{gr}_{\mathfrak{m}} R = \text{Sym}_K(\mathfrak{m}/\mathfrak{m}^2)$, and the chosen derivations span its dual degree-one directions.

If a frame function has order $r < h$, its initial form is a nonzero homogeneous polynomial of degree r . Apply r parameter derivations and reduce modulo $\mathfrak{m}$. Terms in which a derivative hits a coefficient, or fewer than r parameter factors, retain a parameter factor; terms of order at least $r + 1$ do also. The remaining terms pair the initial form with all products of the dual normal directions. In dual coordinates, a derivative of multiorder α gives $\alpha!$ times the corresponding coefficient. Characteristic zero makes $\alpha!$ nonzero. All these derivatives vanish by **Step 2**, a contradiction. The case $r = 0$ is the asserted vanishing of the section on Γ itself.

Every frame function therefore belongs to the localization of I_{Γ}^h . Finitely many denominators suffice for the finite frame, so this containment holds on ambient neighborhoods of a dense open of Γ . No global equality of symbolic and ordinary powers is used. On specialization, the image of I_{Γ} is the defining ideal of the reduced integral fiber V_t , and images of powers are powers of images. The specialized frame spans the entire higher kernel because $\mathcal{K}$ is a subbundle. Hence its evaluation ideal is exactly $\mathfrak{b}_t$ and lies in $I_{V_t}^h$. Intersect with the support-isolation open from **Step 1**; flatness and integrality allow a shrink so these assertions hold generically on general fibers.

Step 4. We restrict the whole image series through an admissible incidence point. Near z , support equality and Noetherianity give, for some $N > 0$,

$$I_{V_t}^N \subset \mathfrak{b}_t \subset I_{V_t}^h.$$

Restrict these inclusions to Z_b . The image of I_{V_t} defines $V_t \times_B \{b\}$, which is the scheme fiber over b in its actual reduced image W_t . Smoothness at z makes this fiber reduced with a unique local

component F . Thus the image ideal is I_{F/Z_b} near z , proving (3.14) and equality of supports. No flatness of $Z_b \rightarrow Z$ is needed.

Restriction further to C gives the reduced ideal of F in C . The restrictions of a full kernel basis generate the base ideal of (3.15); basis vectors restricting to zero do not change that ideal. If the image were zero its support would be all of C , contradicting the local support $F \neq C$. The generic point of F belongs to every nonempty open of F , so the resulting generic base component is isolated and its localized ideal primary.

For the density assertion, form the reduced projective image $\mathcal{W}$ of $\Gamma \rightarrow T \times B$. Its geometric generic fiber over T is integral as the reduced image of the geometrically integral generic fiber of Γ . Spreading and shrinking make its fibers geometrically integral, and projective surjectivity makes them the actual images W_t . Generic smoothness for $\Gamma \rightarrow \mathcal{W}$ and $\mathcal{W} \rightarrow T$, followed by base change to t , gives a dense incidence open with the two stipulated smoothness conditions. Intersect it with Ω . Dominance of evaluation also permits any prescribed very-general condition on z . No arbitrary special fiber was fixed in advance.

Step 5. We verify the stated positivity qualification. If V_t meets the isomorphism open of ϕ , its image has dimension f and positive generic degree e . Projection formula gives

$$m^f P^f \cdot V_t = e A^f \cdot \phi(V_t) > 0.$$

This makes no assertion about subvarieties contracted by the semiample morphism and completes the proof. $\square$

Corollary 3.7. *In Proposition 3.6, assume the ambient is $\mathbb{Q}$ -factorial klt and the generic point of the chosen center is smooth. On a smooth resolution of a general center V of dimension $f > 0$,*

$$(3.16) \quad P^f \cdot V \leq (2/\delta)^{d-f} P^d, \quad K_{V^*} \cdot P^{f-1} \leq (K_Z + cP)|_V \cdot P^{f-1}, \quad 0 < c \leq 2(d-f)/\delta,$$

where powers on V^* mean pulled-back classes. On an integral projective $\mathbb{Q}$ -factorial klt ambient fiber of dimension e , a proper positive-dimensional component F from part (3), generically smooth on that ambient, satisfies the corresponding bounds with d, f, P replaced by $e, \dim F, P|_C$.

Proof. The full higher ideal and its restricted image have the ordinary multiplicity h and generic isolation required by Proposition 3.5. Use (3.9), (3.11), and $k/h \leq 2/\delta$. Proposition 3.4 supplies the generic-subadjunction assertion for both ambients. $\square$

4. A LOWER-DIMENSIONAL BOUND FOR COVERING CURVES

In this section, we prove the consequence of canonical induction that excludes centers with bounded normalized volume and canonical degree.

Theorem 4.1 (Canonical modifications). *Every normal projective variety T admits a projective birational morphism $p: W \rightarrow T$ with W canonical and K_W p -ample. More generally, for a normal pair (T, Δ) with rational coefficients in $[0, 1]$, there exists a projective canonical modification $p: W \rightarrow T$, isomorphic over generic codimension-one points, such that the strict-transform pair (W, Δ_W) is canonical and $K_W + \Delta_W$ is rational Cartier and p -ample. Every coefficient-one component S of Δ_W is normal and canonical, and its different is zero:*

$$(4.1) \quad (K_W + \Delta_W)|_S \sim_{\mathbb{Q}} K_S.$$

The original T need not be $\mathbb{Q}$ -Gorenstein, and the original pair need not be lc.

The variety assertion is [Kol23, author final, Theorem 11.28(1) and Definition 11.29]. The pair assertion and (4.1) are [Fuj15, arXiv v1, Theorem 3.5, Remark 2.4(1), and Proposition 3.6], with the conventions of [Fuj15, arXiv v1, Definitions 2.3 and 2.6]. These are relative canonical models of resolutions, using relative finite generation; they impose no absolute pseudo-effectivity hypothesis.

Theorem 4.2 ([BZ16, Theorem 1.3]). *For each dimension f there exists an integer $m_f > 0$ such that, for every smooth projective f -fold Y and every nef integral Cartier divisor N with $K_Y + N$ big, the system $|m_f(K_Y + N)|$ is birational. The integer is independent of N .*

This is the specialization with boundary zero and Cartier index one of the polarized effective birationality theorem.

Theorem 4.3 (Conditional exclusion of centers). *Assume CS_f , where $f \geq 1$. Let X be normal projective with a strictly nef Cartier divisor L , and let $g: Y \rightarrow X$ be a morphism from a smooth integral projective f -fold, generically finite onto its positive-dimensional image. Put $M = g^*L$. Then $K_Y + bM$ is big for every rational $b > 2f$.*

Fix an integer $b > 2f$. With m_f as in Theorem 4.2, let Q be nef rational Cartier and let $R, C_0 > 0$, $C \geq 0$, satisfy

$$(4.2) \quad 0 < Q^f \leq C_0, \quad Q - RM \text{ is nef}, \quad K_Y \cdot Q^{f-1} \leq CQ^f.$$

Then an algebraic covering family of integral curves on Y has general member C_Y satisfying

$$(4.3) \quad R \leq Q \cdot C_Y \leq \left[m_f \left(C + \frac{b}{R} \right) \right]^{f-1} C_0.$$

For $f = 1$, the exponent zero means one and the curve is Y itself. Thus such configurations cannot exist with $R \rightarrow \infty$ and f, b, C_0, C bounded. No singularity condition or movement condition is imposed on $g(Y)$ or its normalization.

Proof. We first modify the normal finite factor of g , so that the canonical divisor is positive on curves where the pulled-back polarization vanishes. We then return to the smooth Y for the effective birationality argument.

Step 1. We obtain a strictly nef adjoint on a canonical modification. Let $V = g(Y)_{\text{red}}$ and take Stein factorization $Y \xrightarrow{\mu} T \xrightarrow{h} V$. The first map is birational because g is generically finite and μ has connected generic fiber. The normal projective T has strictly nef Cartier divisor $M_T = h^*(L|_V)$: a finite map sends an integral curve to a curve and multiplies its positive degree by a positive integer. We have $M = \mu^*M_T$.

Take the canonical modification $p: W \rightarrow T$ of Theorem 4.1, and put $M_W = p^*M_T$. A curve has M_W -degree zero exactly when it is contracted by p , and then its K_W -degree is positive. On a K_W -negative extremal rational curve, Theorem 2.2 gives length at most $2f$. Such a curve is not contracted by p , so its integral M_W -degree is at least one. Hence $K_W + 2fM_W$ is nef. For rational $b > 2f$, the divisor $F = K_W + bM_W$ is strictly nef: on noncontracted curves use $(b - 2f)M_W$, and on contracted curves use K_W .

Choose $\ell > 0$ such that ℓF is Cartier. By CS_f , $K_W + s\ell F$ is big for some rational $s > 0$. The identity

$$(4.4) \quad (1 + s\ell)F = (K_W + s\ell F) + bM_W$$

makes F big. The index ℓ is used only in this qualitative application.

Step 2. We descend bigness to the smooth center Y . Choose a common smooth resolution with $a: U \rightarrow Y$ and $c: U \rightarrow W$ over T . Canonicity gives $K_U = c^*K_W + E_W$ with $E_W \geq 0$ exceptional, so

$$K_U + ba^*M = c^*F + E_W$$

is big. Also $K_U = a^*K_Y + E_Y$ with $E_Y \geq 0$ exceptional. For sufficiently divisible k , $a_*\mathcal{O}_U(kE_Y) = \mathcal{O}_Y$: a rational function with poles allowed only along exceptional primes has no pole at a codimension-one point of the normal Y . Projection formula identifies the sections of $k(K_U + ba^*M)$ with those of $k(K_Y + bM)$. Their growth proves bigness of $K_Y + bM$.

Step 3. We bound degrees using a fixed Cartier polarization. For integral $b > 2f$, Theorem 4.2 applies to the nef Cartier divisor bM , with index one independent of b and M . Resolve the base ideal of $|m_f(K_Y + bM)|$ by $v: Y' \rightarrow Y$ and write

$$v^*(m_f(K_Y + bM)) = H + E,$$

where H is the basepoint-free integral moving part, $E \geq 0$, and the morphism of H is birational onto its image. In particular H is nef and big. Write $Q' = v^*Q$ and $M' = v^*M$. From (4.2), $M \cdot Q^{f-1} \leq Q^f/R$, and hence

$$(4.5) \quad H \cdot (Q')^{f-1} \leq m_f(K_Y + bM) \cdot Q^{f-1} \leq m_f(C + b/R)Q^f.$$

The effective fixed part has nonnegative intersection with $(Q')^{f-1}$.

For $f \geq 2$, the positive mixed numbers $s_i = (Q')^{f-i}H^i$ satisfy $s_i^2 \geq s_{i-1}s_{i+1}$ by [Laz04, Corollary 1.6.3(i) and Example 1.6.4]. Their successive ratios decrease. Thus

$$(4.6) \quad Q' \cdot H^{f-1} \leq Q^f \left(\frac{H \cdot (Q')^{f-1}}{Q^f} \right)^{f-1} \leq [m_f(C + b/R)]^{f-1}C_0.$$

Step 4. We construct actual covering curves with positive integral degree downstairs. Choose y' where v and the morphism defined by H are isomorphisms onto their images and $g \circ v$ is quasi-finite. Cut its image by $f - 1$ general hyperplanes through the image of y' . Their pullbacks intersect properly: the subsystem is generated away from y' , and their local equations can be independent at y' . The resulting effective cycle has class H^{f-1} and an integral component B' through y' . Nefness gives $Q' \cdot B' \leq Q' \cdot H^{f-1}$. Its image $C_Y = v(B')$ has mapping degree one, and $g(C_Y)$ is a curve because $g \circ v$ is quasi-finite at y' . Therefore $M \cdot C_Y \geq 1$, and

$$Q \cdot C_Y \geq RM \cdot C_Y \geq R.$$

Together with (4.6) this proves (4.3).

Varying the hyperplanes gives an algebraic family of complete intersections. The finitely many component families of its generic member, after finite parameter extension and shrinking when needed, include a family whose evaluation dominates the isomorphism open of Y' . Pushing it down gives the required covering family on Y . For $f = 1$, the nonconstant map gives $M \cdot Y \geq 1$, and $R \leq Q \cdot Y \leq C_0$ directly. Finally the upper bound in (4.3) is at most $[m_f(C + b)]^{f-1}C_0$ for $R \geq 1$, proving the final assertion. $\square$

5. A SCALAR BOUND FOR COMPLETE JET KERNELS

In this section, we prove the single-point vanishing bound by applying Theorem 4.3 to the centers furnished by Proposition 3.6. This replaces the canonical positivity input in the scalar construction of [Ope26b, Lemma 5.2] by the stated canonical induction hypothesis.

Theorem 5.1 ([BCHM10, Corollary 1.4.3]). *A quasi-projective klt rational pair (X, B) has a small projective birational morphism $\pi: X' \rightarrow X$ with X' $\mathbb{Q}$ -factorial, effective strict transform B' , and*

$$(5.1) \quad K_{X'} + B' = \pi^*(K_X + B),$$

with (X', B') klt. Here small means that π has no exceptional prime divisor. The morphism is an isomorphism over X_{reg} .

This is the empty-extraction specialization of the cited corollary. Its relative construction contracts the exceptional divisors not prescribed for extraction; it requires no absolute pseudo-effectivity. Over a locally factorial open of X , a projective small morphism is an isomorphism: a relatively ample divisor and the pullback of its pushforward have only exceptional divisorial difference, which is zero. In particular it is an isomorphism over X_{reg} .

Theorem 5.2 (Scalar vanishing bound). *Assume $n \geq 2$ and $\text{CS}_{<n}$. Consider either a smooth projective n -fold X , with $B = 0$ and $\tau = n + 1$, or an arbitrary projective klt rational pair (X, B) , with $\tau = 2n$. Let L be strictly nef Cartier, A ample Cartier, $t > \tau$ rational, $D = K_X + B + tL$, and $\lambda = t - \tau$. Fix $0 < v_- \leq v_+$. There exists r_0 , depending only on n, λ, v_-, v_+ , such that for positive rational u, r with $r \geq r_0$ and $P = r(D + uA)$ satisfying $v_- \leq P^n \leq v_+$, every smooth point outside a countable union of proper closed subsets satisfies*

$$(5.2) \quad \text{ord}_x(s) \leq 4k(P^n)^{1/n}$$

for every Cartier kP and every nonzero $s \in H^0(X, kP)$. No $\mathbb{Q}$ -factoriality or separate $\mathbb{Q}$ -Cartierness of K_X is required in the pair setting.

Explicitly, with $b = 2n + 1$, $\delta_* = 2v_-^{1/n}/n$, and m_f from Theorem 4.3, one may take

$$(5.3) \quad \begin{aligned} B_f &= (2/\delta_*)^{n-f} v_+, & T_f &= \left[m_f \left(\frac{2(n-f)}{\delta_*} + 1 + \frac{b}{\lambda} \right) \right]^{f-1} B_f, \\ r_0 &= 2 + \frac{\max_{1 \leq f < n} T_f}{\lambda}. \end{aligned}$$

A common very-general locus works for any countable collection of such normalizations. This includes smooth and $\mathbb{Q}$ -factorial canonical ambients directly and arbitrary canonical ambients via the pair setting.

Proof. The smooth setting is used on X itself. In the pair setting we take a small model only to apply the local numerical tools; the target of the center maps remains the original X .

Step 1. We record the inherited nef comparisons and section spaces. Proposition 2.3 makes $K_X + B + \tau L$ nef. Thus D is nef, P is ample rational Cartier, and

$$(5.4) \quad P - r\lambda L = r(K_X + B + \tau L + uA) \text{ is nef.}$$

In the pair setting use $\pi: X' \rightarrow X$ from Theorem 5.1, and put $M' = \pi^*L$, $A' = \pi^*A$, $P' = \pi^*P$. Then P' is nef, big, and semiample, and

$$(5.5) \quad P' - r\lambda M' \text{ is nef,} \quad P'/r - K_{X'} = B' + tM' + uA'.$$

We do not assert nefness of the effective term B' . Choose a dense smooth open $U \subset X$ over which π is an isomorphism and which is disjoint from $\text{Supp } B$. Normality and projection formula give

$$(5.6) \quad H^0(X', kP') = H^0(X, kP)$$

for every Cartier kP , with equal orders over U . For smooth X , take $X' = X$, $B' = 0$, and π the identity; all comparisons remain valid with $\tau = n + 1$.

Step 2. We convert a violation into full kernels at $n + 1$ levels. Remove the closures of all proper rank-drop loci of section-to-jet maps on X_{reg} , for every admissible integer degree and jet order, and remove $X \setminus U$. This is a countable union E_P of proper closed sets. A kernel nonzero at a point outside it is generically nonzero. If (5.2) fails for s_0 of degree k_0 , put $a_0 = \text{ord}_x(s_0)$ and $\rho = (P^n)^{1/n}$, so $a_0/k_0 > 4\rho$. Powers of generically nonzero kernel members give nonzero kernels at order Na_0 in degree $k = Nk_0$.

Set $\delta = 2\rho/n$ and $\tau_i = 3\rho/2 + i\delta$, $0 \leq i \leq n$. The last level is $7\rho/2 < 4\rho$. Choose N so large that $k\delta \geq 4$ and $Na_0 \geq \lceil k\tau_n \rceil$. By (5.6), all full kernels on X' at these levels are nonzero at general marks. If the lowest base support had an isolated marked point, Proposition 3.5 with $h_0 = \lceil k\tau_0 \rceil$ would give

$$1 \leq (k/h_0)^n (P')^n \leq (2/3)^n < 1.$$

Its base support is proper because the kernel is nonzero. Proposition 3.6 therefore supplies a dominant family of proper positive-dimensional centers.

Step 3. We verify the canonical degree bound on each actual center. The variety X' is klt because (X', B') is klt with effective boundary. A general center V has smooth generic point,

meets $\pi^{-1}(U)$, and is not contained in B' , by incidence dominance. Its semiample image has dimension $f = \dim V$, so its degree is positive. On a smooth resolution $g: Y \rightarrow V \subset X'$ put $Q = g^*P'$, $M = g^*M'$, and $h = \pi \circ g: Y \rightarrow X$. The map h is generically finite onto its f -dimensional image, with $1 \leq f < n$. Corollary 3.7 gives

$$0 < Q^f \leq (2/\delta)^{n-f} P^n \leq B_f, \quad K_Y \cdot Q^{f-1} \leq g^*(K_{X'} + cP') \cdot Q^{f-1}, \quad 0 < c \leq 2(n-f)/\delta.$$

The restriction of B' is effective, since V is not contained in it and X' is $\mathbb{Q}$ -factorial. Testing (5.5) with Q^{f-1} therefore gives

$$(5.7) \quad g^*K_{X'} \cdot Q^{f-1} \leq Q^f/r, \quad K_Y \cdot Q^{f-1} \leq (c + 1/r)Q^f, \quad Q - r\lambda M \text{ is nef.}$$

Only the nonnegative testing intersection of B' is used.

Step 4. We contradict the lower-dimensional curve bound uniformly. Use Theorem 4.3 for $h: Y \rightarrow X$, with $R = r\lambda$, $C_0 = B_f$, $C = c + 1/r$, and the fixed Cartier polarization bM , $b = 2n + 1 > 2f$. Since $\delta \geq \delta_*$ and $r \geq 1$, it gives

$$r\lambda \leq [m_f(c + 1/r + b/(r\lambda))]^{f-1} B_f \leq T_f.$$

But (5.3) makes $r\lambda > T_f$ for every $f < n$. This proves (5.2). The constants involve no center index, small model, or boundary coefficient. Taking the union of the sets E_P for a countable collection preserves a nonempty very-general complement over $\mathbb{C}$. In a family with $r \rightarrow \infty$ and a fixed positive volume window, the bound thus holds eventually. All denominator choices enter only the degrees k , never the constants. $\square$

6. DIVISORS AND VECTOR FIELDS ON CANONICAL COVERS

In this section, we exclude bounded-degree divisors on finite covers and treat the canonical varieties whose tangent sheaves are generically generated by global sections.

Definition 6.1. A finite surjective morphism $\tau: T \rightarrow X$ of normal integral varieties is *quasi-etale* if it is etale away from a closed subset of codimension at least two in X .

Lemma 6.2. *Let X be normal projective canonical and $\tau: T \rightarrow X$ a finite normal quasi-etale cover. Then T is canonical and $K_T = \tau^*K_X$ for compatible canonical divisors. If hK_X is Cartier, then hK_T is Cartier. Finite pullback preserves strict nefness of Cartier divisors.*

Proof. The etale canonical equality holds at all generic divisorial points, since a finite map sends a prime divisor to a prime divisor. It extends as a Weil-divisor equality and proves the Cartier-index assertion. Choose a resolution $w: W \rightarrow X$ with $K_W = w^*K_X + E_X$, $E_X \geq 0$. Resolve the normalization of the dominant component of $T \times_X W$ to obtain $a: Y \rightarrow T$ birational and $b: Y \rightarrow W$ generically finite, with Y smooth. The determinant of the differential gives an effective ramification divisor R_b , and

$$K_Y - a^*K_T = b^*E_X + R_b \geq 0.$$

This is the canonical discrepancy condition. For a curve $C \subset T$, finite pullback has degree $\deg(C \rightarrow \tau(C))L \cdot \tau(C)$, which is positive when L is strictly nef. $\square$

Proposition 6.3 (Uniform exclusion of bounded divisors). *Assume $n \geq 2$ and CS_{n-1} . Fix $\lambda, v_-, B_0 > 0$. There exists r_0 , depending only on these numbers and n , such that the following holds. Let X be projective canonical of dimension n , let L be strictly nef Cartier, and let P be ample rational Cartier with*

$$(6.1) \quad P^n \geq v_-, \quad P - r\lambda L \text{ nef}, \quad P/r - K_X \text{ nef}, \quad r \geq r_0.$$

For every finite normal quasi-etale cover $\tau: T \rightarrow X$ and every prime Weil divisor J on T ,

$$(6.2) \quad J \cdot (\tau^*P)^{n-1} > B_0.$$

This also excludes nonzero effective integral Weil divisors of degree at most B_0 , with no bound on the cover degree or component index. One may take, with $f = n - 1$ and $b = 2n + 1$,

$$(6.3) \quad C_B = [m_f(1 + B_0/v_- + b/\lambda)]^{n-2} B_0, \quad r_0 = 2 + C_B/\lambda.$$

The exponent zero has value one.

Proof. We use a small model to make a prime rational Cartier, and a canonical modification to obtain a divisorial adjunction inequality without assuming normality or lc singularities of that prime.

Step 1. We pass to a crepant small model of the cover. By Lemma 6.2, T is canonical with the canonical index inherited from X . Theorem 5.1 gives a small projective $\sigma: T' \rightarrow T$ with T' $\mathbb{Q}$ -factorial and canonical. Suppose a prime J has degree $0 < d \leq B_0$. Let J' be its strict transform, and set $P' = \sigma^* \tau^* P$, $M' = \sigma^* \tau^* L$, $e = \deg \tau$. Then

$$(6.4) \quad (P')^n = eP^n, \quad J' \cdot (P')^{n-1} = d, \quad P' - r\lambda M' \text{ nef}, \quad P'/r - K_{T'} \text{ nef}.$$

The effective prime J' is rational Cartier; its index will not enter a bound.

Step 2. We compare canonical degree on a resolution of the prime. Take the canonical modification $p: Z \rightarrow T'$ of (T', J') from Theorem 4.1. Its strict transform S is normal canonical and $(K_Z + S)|_S \sim_{\mathbb{Q}} K_S$. Put

$$E = p^*(K_{T'} + J') - (K_Z + S).$$

It is exceptional and $-E$ is relatively ample. The negativity lemma [KM98, Lemma 3.39] (see also [Bir07, Lemma 3.0.45]) gives $E \geq 0$. Since S is not exceptional it is not contained in E , so an effective Cartier multiple restricts nontrivially to S and $E|_S \geq 0$. With $P_S = (p^*P')|_S$,

$$K_S + E|_S \sim_{\mathbb{Q}} (p^*(K_{T'} + J'))|_S.$$

Resolve $v: Y \rightarrow S$ and put $Q = v^*P_S$. Canonicity gives $K_Y = v^*K_S + F$, $F \geq 0$ exceptional. Every component of F has image of dimension at most $n - 3$, so $F \cdot Q^{n-2} = 0$. For $n = 2$, the normal curve S is smooth and $F = 0$. Projection formula and the effective restriction give

$$(6.5) \quad K_Y \cdot Q^{n-2} \leq (K_{T'} + J') \cdot J' \cdot (P')^{n-2}.$$

No pullback of the canonical divisor of an arbitrary normalization of J' occurs.

Step 3. We bound the canonical ratio independently of indices and cover degree. The nef comparison in (6.4) and the effective prime cycle give $K_{T'} \cdot J' \cdot (P')^{n-2} \leq d/r$. Hodge index gives

$$(J')^2 \cdot (P')^{n-2} \leq \frac{d^2}{(P')^n}.$$

For this use on a singular small model, pull both rational Cartier classes to a smooth resolution and approximate the nef big pullback of P' by ample classes. Apply the ample Hodge inequality and pass to the limit, whose denominator stays positive; see [Laz04, Section 1.6]. Thus

$$(6.6) \quad K_Y \cdot Q^{n-2} \leq d/r + d^2/(eP^n) \leq (1/r + B_0/v_-)d.$$

Also $Q^{n-1} = d$. The composite $g: Y \rightarrow X$ is generically finite onto the prime image of J , and $Q - r\lambda g^*L$ is nef. Theorem 4.3, with $f = n - 1$, gives

$$r\lambda \leq [m_f(1/r + B_0/v_- + b/(r\lambda))]^{n-2} B_0 \leq C_B$$

for $r \geq 2$. This contradicts (6.3). A nonzero effective integral Weil divisor has a prime component with positive integer multiplicity and no larger degree, proving the additional conclusion. $\square$

Corollary 6.4. Assume CS_{n-1} . Fix canonical X , strictly nef Cartier L , ample Cartier A , and rational $t > 2n$. If $P_j = r_j(K_X + tL + u_j A)$, with positive real $r_j \rightarrow \infty$, $u_j > 0$, and $\liminf P_j^n > 0$, then the minimum prime-divisor degree on every finite normal quasi-etale cover tends to infinity uniformly in the cover. No upper volume bound or nonbigness assumption is needed.

Proof. Put $D = K_X + tL$, $\lambda = t - 2n$. Proposition 2.3 gives D nef and the two comparisons in (6.1). For real coefficients, choose $v > 0$ with $P_j^n \geq v$ eventually and rationals $r_j/2 < s_j < r_j$, $0 < a_j < u_j$, sufficiently close to r_j, u_j that $\tilde{P}_j = s_j(D + a_j A)$ has volume at least $v/2$. Then

$$P_j - \tilde{P}_j = (r_j - s_j)D + (r_j u_j - s_j a_j)A$$

is nef. Proposition 6.3 applies to $\tilde{P}_j$ at scale $s_j \rightarrow \infty$. For every prime cycle on a cover, the factorization

$$P_j^{n-1} - \tilde{P}_j^{n-1} = (P_j - \tilde{P}_j) \sum_{a=0}^{n-2} P_j^{n-2-a} \tilde{P}_j^a$$

has nonnegative intersection after pullback. The asserted uniform bound follows for P_j . $\square$

Theorem 6.5 ([GKKP11, Theorem 6.1]). *A complex klt variety with locally free tangent sheaf is smooth.*

Theorem 6.6 ([MU19, Theorem 4.14(iv) $\Rightarrow$ (i)]). *A smooth projective variety with trivial cotangent bundle is an abelian variety after choosing an origin.*

Theorem 6.7 ([Ser95, Proposition 1.4]). *A strictly nef Cartier divisor on an abelian variety is ample.*

Proposition 6.8 (Generic tangent generation). *Assume $n \geq 2$ and CS_{n-1} . Let Y be a normal projective canonical n -fold whose reflexive tangent sheaf $T_Y = \mathcal{H}om_{\mathcal{O}_Y}(\Omega_Y^1, \mathcal{O}_Y)$ is generically generated by global sections. If M is strictly nef Cartier, then $K_Y + bM$ is ample for every real $b > 2n$. If $\tau: Y \rightarrow X$ is a finite normal quasi-etale cover of canonical X and $M = \tau^*L$ for strictly nef Cartier L , the corresponding conclusion holds on X . No CS_n hypothesis is used.*

Proof. Choose n global vector fields independent at a smooth point in the generation open. Their wedge defines an effective integral Weil divisor $G \sim -K_Y$ by closure from Y_{reg} , or equivalently a section of $\mathcal{O}_Y(-K_Y)$. There are no divisorial poles, and G is rational Cartier since K_Y is. We treat its nonzero and zero alternatives.

Step 1. We obtain a canonical component when $G \neq 0$. Write $G = \sum m_i S_i$ with positive integers m_i , put $m = \max m_i$, $\epsilon = 1/m$, and $\lambda = 1 - \epsilon \in [0, 1)$. The rational boundary $\Theta = \epsilon G$ has a coefficient-one component S and

$$(6.7) \quad K_Y + \Theta \sim_{\mathbb{Q}} \lambda K_Y.$$

The sign $\lambda \geq 0$ uses integrality of G .

Take the canonical modification $f: Z \rightarrow Y$ of (Y, Θ) . By Theorem 4.1, the strict component S_Z is normal canonical with zero different. The exceptional divisor

$$E = f^*(K_Y + \Theta) - (K_Z + \Theta_Z)$$

has $-E$ relatively nef, so $E \geq 0$ by [KM98, Lemma 3.39] (see also [Bir07, Lemma 3.0.45]). The component S_Z is not exceptional and its restriction E_S is effective. Put $M_S = (f^*M)|_{S_Z}$. Restricting (6.7) gives

$$(6.8) \quad (f^*(\lambda K_Y))|_{S_Z} \sim_{\mathbb{Q}} K_{S_Z} + E_S.$$

Step 2. We use only lower-dimensional induction on the transformed component. For a smooth resolution $v: W \rightarrow S_Z$, the composite $W \rightarrow Y$ is generically finite onto the prime divisor S . Theorem 4.3 gives $K_W + uv^*M_S$ big for every rational $u > 2(n-1)$. Since S_Z is canonical, $K_W = v^*K_{S_Z} + F$ with effective exceptional F . Normality gives $v_*\mathcal{O}_W(kF) = \mathcal{O}_{S_Z}$ for divisible k , because allowed exceptional poles give no divisorial poles downstairs. Equality of sections then proves $K_{S_Z} + uM_S$ big. Adding E_S to (6.8) proves

$$(6.9) \quad (f^*(\lambda K_Y + uM))|_{S_Z} \text{ is big} \quad (u > 2(n-1), u \in \mathbb{Q}).$$

This does not require M_S itself to be strictly nef.

Step 3. We contradict a nonample adjoint in both coefficient cases. If one adjoint above $2n$ fails to be ample, Proposition 2.3 gives $K_Y^i M^{n-i} = 0$ for all $0 \leq i \leq n$. For rational $s > 2n$, $N_s = K_Y + sM$ is nef, and

$$G \cdot N_s^{n-1} = -K_Y \cdot (K_Y + sM)^{n-1} = 0, \quad G \cdot M^{n-1} = 0.$$

All component intersections are nonnegative and all $m_i > 0$, so $S \cdot N_s^{n-1} = S \cdot M^{n-1} = 0$. If $\lambda > 0$, choose rational $u > \max\{2(n-1), 2n\lambda\}$ and set $s = u/\lambda$. By (6.9), $(f^*N_s)|_{S_Z}$ is nef and big, yet its top intersection equals $N_s^{n-1} \cdot S = 0$, a contradiction. If $\lambda = 0$, (6.9) makes M_S nef and big, whereas $M_S^{n-1} = M^{n-1} \cdot S = 0$. This includes the reduced anticanonical divisor case and proves the assertion for $G \neq 0$.

Step 4. We extend the vector-field frame when $G = 0$. On $U = Y_{\text{reg}}$ the wedge has no zero: locally a nonzero function defining a section of a line bundle, if it vanishes, defines a nonempty Cartier divisor with a codimension-one component. Thus the evaluation $\mathcal{O}_U^n \rightarrow T_U$ is an isomorphism everywhere. Both sheaves extend reflexively across $Y \setminus U$, of codimension at least two. Pushing the map and its inverse by $j: U \hookrightarrow Y$ gives

$$T_Y \cong \mathcal{O}_Y^n$$

on all of Y . Theorem 6.5 now makes Y smooth, so Theorem 6.6 applies and makes it abelian. Theorem 6.7 makes M ample. Its numerical mechanism is that the connected kernel of the homomorphism $a \mapsto t_a^* \mathcal{O}(M) \otimes \mathcal{O}(-M)$ is an abelian subvariety on which M is numerically trivial. Strict nefness excludes positive dimension; the homomorphism is an isogeny and the abelian Riemann–Roch degree formula gives $(M^n/n!)^2 > 0$. Since $K_Y \sim 0$, this also yields positivity by Proposition 2.3.

Step 5. We descend through the finite cover. Lemma 6.2 gives $K_Y + bM = \tau^*(K_X + bL)$. For rational $b > 2n$, ampleness descends by the finite-map criterion after taking a Cartier multiple. Interpolation between rational parameters proves it for real b . A regular connected algebraic group action with open orbit supplies the hypothesis on T_Y : at a smooth orbit point the characteristic-zero orbit map is separable and its differential spans the tangent space, while its infinitesimal action consists of global derivations. This observation presupposes a regular action; no birational-group assertion has entered the argument. $\square$

7. JETS ON A PROJECTIVE BUNDLE OVER TWO FACTORS

In this section, we prove the jet-generation statement used in the determinant argument, with the fiber coefficient fixed before the normalization parameter grows.

Set-up 7.1. Let X be normal projective canonical of dimension $n \geq 2$, let L be strictly nef Cartier, let A be ample Cartier, and let $t > 2n$ be rational. Put $D = K_X + tL$ and $\lambda = t - 2n > 0$. Assume D is not big, and let $u_j > 0$ be rational and $r_j > 0$ integral, with $u_j \rightarrow 0$, $r_j \rightarrow \infty$, such that $P_j = r_j(D + u_j A)$ satisfies $P_j^n \rightarrow \epsilon^n > 0$.

Write L_i and $P_{j,i}$ for pullbacks from the two factors of $X \times X$, and set

$$(7.1) \quad \pi: Z = \mathbb{P}_{X \times X}(\mathcal{O}_X(L)_1 \oplus \mathcal{O}_X(L)_2) \rightarrow X \times X, \quad \xi = c_1(\mathcal{O}_Z(1)).$$

We use the quotient convention $\pi_* \mathcal{O}_Z(a) = \text{Sym}^a(\mathcal{O}_X(L)_1 \oplus \mathcal{O}_X(L)_2)$ for $a \geq 0$. The two coordinate sections vanish on the Cartier divisors S_i of class $\xi - L_i$, called the *axes*. Their complement is the *torus open*. For $q > 0$, put $R_j = P_{j,1} + P_{j,2} + q\xi$. For a smooth point z , define

$$(7.2) \quad \epsilon(R_j; z) = \inf_{C \ni z} \frac{R_j \cdot C}{\text{mult}_z C},$$

where C ranges over integral curves through z .

Lemma 7.2 (Numerical estimates after a small modification). *Let W be projective klt, with rational Cartier K_W , and let H be nef, big, and semiample rational Cartier. Form complete jet kernels on W at smooth marks. If a moving center V from Proposition 3.6 meets W_{reg}*

and has $H^{\dim V} \cdot V > 0$, the degree and canonical estimates in (3.16) hold without assuming W $\mathbb{Q}$ -factorial. The corresponding assertion holds for a proper component obtained by restricting the whole higher kernel to an integral projective klt ambient fiber, smooth at the generic point of the component.

Proof. Take a small model $\sigma: W' \rightarrow W$ from Theorem 5.1. It is crepant and isomorphic over W_{reg} . Normality identifies complete section spaces, hence the kernels and generic base ideals at those marks. The pullback of H is nef, big, and semiample. The strict transform of V meets the isomorphism open, has unchanged degree, and satisfies the ordinary-ideal and isolation hypotheses of Proposition 3.5. Proposition 3.4 applies on W' . Projection formula gives the asserted inequalities on a common resolution of V ; additional exceptional canonical divisors on a smooth resolution have zero intersection with the pulled-back testing power. For a fiber, first restrict the original higher kernel as in (3.15), and then take a separate small model of that fiber. No commutation of small modifications with fibers is required. $\square$

Lemma 7.3 (Finiteness over a fixed smooth open). *Let X be normal complex projective and let $E > 0$ be an integer. Up to isomorphism over X , only finitely many connected finite normal covers of degree at most E are étale over X_{reg} .*

Proof. A finite semialgebraic triangulation of the compact X compatible with its singular locus makes X_{reg} homotopy equivalent to a finite complex. Its topological fundamental group is finitely generated, so it has finitely many permutation representations in each symmetric group of degree at most E . Riemann existence [Gro03, Exposé XII, Theorem 5.1] gives finitely many corresponding finite étale covers of X_{reg} . Normalization of X in each function field is its unique normal finite completion. $\square$

Theorem 7.4 (Regular automorphisms, [Bla17, Introduction and Section 2.1]). *For normal projective varieties, isomorphisms are represented by the open graph subfunctor of the Hilbert scheme of their product. This scheme is a countable union of schemes of finite type. The identity component of the regular automorphism scheme in characteristic zero is an algebraic group; its other components are translates. Its action is regular, and differentiation gives global derivations.*

Theorem 7.5 (Two-slot jets). *Assume $\text{CS}_{<n}$ and use Set-up 7.1. For each real $h > 0$ there exist an integer $q > 0$, independent of j , and j_0 such that, for every $j \geq j_0$ and every pair of nonempty opens $U_1, U_2 \subset X_{\text{reg}}$, some torus point z over $U_1 \times U_2$ satisfies $\epsilon(R_j; z) > h$.*

For each such z and each positive rational $a < \epsilon(R_j; z)$, $|kR_j|$ generates jets through order ka for every sufficiently large divisible k with kR_j Cartier and ka integral. Thus $h = 2n + 2$ gives one fixed q and complete jets through order $(2n + 2)k$. No scalar jet hypothesis is required, and the assertion applies without $\mathbb{Q}$ -factoriality of X .

Proof. We exclude a dominant family of common jet-base components on the original bundle Z . All maps from centers and covers retain the original variety X as target. Constants below are independent of j , the prescribed opens, and the centers or covers chosen at that index.

Step 1. We fix the numerical comparisons and the geometry of the slices. Write P, r, u for P_j, r_j, u_j and take the eventual volume window $v_- = \epsilon^n/2 \leq P^n \leq 2\epsilon^n = v_+$. Proposition 2.3 gives

$$(7.3) \quad P - r\lambda L \text{ nef}, \quad P/r - K_X = tL + uA \text{ nef}, \quad -K_X = tL - D.$$

The divisor D is nef. On any smooth center of dimension $f < n$, Theorem 4.3 with $b = 2n + 1$ excludes, as $r \rightarrow \infty$, simultaneous bounds

$$(7.4) \quad 0 < Q^f \leq B, \quad Q - r\lambda g^*L \text{ nef}, \quad K_Y \cdot Q^{f-1} \leq cQ^f$$

for fixed $B > 0, c \geq 0$, whenever $g: Y \rightarrow X$ is generically finite onto its image. Fixed images are allowed. Proposition 6.3 also excludes every nonzero effective integral Weil divisor of bounded P -degree on every quasi-étale cover, uniformly in the cover.

The class ξ is nef, since both summands are nef line bundles, and is relatively ample. The class $R = P_1 + P_2 + q\xi$ is ample: for large m , $\xi + m(P_1 + P_2)$ is ample, and

$$R = m^{-1}(\xi + m(P_1 + P_2)) + (q - m^{-1})\xi$$

is ample when $m > 1/q$. The product and bundle are canonical: on a product of resolutions and the pulled-back bundle their discrepancies are sums of pullbacks of the effective discrepancies of X . Their canonical classes are

$$(7.5) \quad K_Z = K_{X,1} + K_{X,2} - S_1 - S_2.$$

A fiber over a smooth point of either projection $Z \rightarrow X$ is the integral projective canonical $(n+1)$ -fold

$$(7.6) \quad p: Z_{\text{sl}} = \mathbb{P}_X(\mathcal{O}_X \oplus \mathcal{O}_X(L)) \rightarrow X, \quad Q_{\text{sl}} = P + q\xi, \quad K_{Z_{\text{sl}}} = p^*K_X - 2\xi + p^*L.$$

Its axes have classes ξ and $\xi - L$, and it is smooth over X_{reg} . These are the ambients to which Lemma 7.2 will be applied separately.

Step 2. We choose the gaps and the fiber coefficient before taking large j . Put $d = 2n + 1$. For $r\lambda \geq q$, projective-bundle intersection gives

$$(7.7) \quad d \binom{2n}{n} q v_-^2 \leq R^d \leq 3^d q v_+^2, \quad Q_{\text{sl}}^{n+1} \leq (2^{n+1} - 1) q v_+.$$

To check the bounds, $\pi_*1 = 0$ and

$$\pi_*\xi^a = \sum_{i=0}^{a-1} L_1^i L_2^{a-1-i} \quad (a \geq 1).$$

The term with one ξ is $d \binom{2n}{n} q (P^n)^2$. Replacing each L by $P/(r\lambda)$ in mixed nef intersections bounds the term with a copies by

$$\binom{d}{a} 2^{d-a} q \left(\frac{q}{r\lambda}\right)^{a-1} (P^n)^2.$$

Summing gives the first upper bound. On the slice, $p_*\xi^a = L^{a-1}$ gives the second. These are Cartier Chow intersections and require no $\mathbb{Q}$ -factoriality.

Choose a positive rational δ with

$$(7.8) \quad (2/\delta)^n (2^{n+1} - 1) v_+ < 1.$$

Take positive rational $\tau_0 > h$ and $\tau_i = \tau_0 + i\delta$ for $0 \leq i \leq d$. Then choose an integer $q > \tau_d$ so large that

$$(7.9) \quad \tau_d^d < d \binom{2n}{n} q v_-^2.$$

All these choices precede j .

Step 3. We turn failure of the Seshadri assertion into moving centers. Suppose failures occur at arbitrarily large indices, with possibly varying opens U_1, U_2 . At a fixed such index, every smooth torus point over their product has Seshadri constant at most h . Asymptotic Riemann–Roch for ample rational R gives $h^0(Z, kR) = R^d k^d / d! + O(k^{d-1})$. Jets of order less than $\lceil k\tau_d \rceil$ have length $\tau_d^d k^d / d! + O(k^{d-1})$. By (7.7) and (7.9), all $d+1$ full kernels are nonzero for large divisible k .

At each such point z , choose an integral curve $C \ni z$ with $R \cdot C / \text{mult}_z C < \tau_0$. Every section in the lowest kernel vanishes on C : otherwise its local intersection is at least $\lceil k\tau_0 \rceil \text{mult}_z C > kR \cdot C$. Thus the lowest base support has a positive-dimensional component through z . For $k\delta \geq 4$,

Proposition 3.6 and Lemma 7.2 give moving integral centers V , $1 \leq f \leq 2n$, with dominant incidence, and on a resolution

$$(7.10) \quad \begin{aligned} 0 &< R^f \cdot V \leq (2/\delta)^{d-f} R^d, \\ K_{V^*} \cdot R^{f-1} &\leq (K_Z + cR)|_V \cdot R^{f-1}, \quad 0 < c \leq 2(d-f)/\delta. \end{aligned}$$

They meet the smooth torus open, so their pulled-back axes are effective. Since $R/r - (K_{X,1} + K_{X,2})$ is nef, (7.5) implies

$$(7.11) \quad K_{V^*} \cdot R^{f-1} \leq (c + 1/r)R^f \cdot V.$$

All degrees and coefficients are uniformly bounded.

Step 4. We exclude all positive intermediate dimensions in the slice. For either projection of V to X , choose the fiber component F through a general incidence point satisfying Proposition 3.6(3). Require both coordinates to be smooth and the point to avoid both axes. If $1 \leq l = \dim F \leq n$, the component is proper in the integral klt slice Z_{sl} . The restricted series is the nonzero image of the entire higher ambient kernel; it has ordinary multiplicity $\lceil k\delta/2 \rceil$ and generic isolation. Its generic point is smooth on the slice. After taking a separate small model of this slice, we obtain

$$(7.12) \quad \begin{aligned} 0 &< Q_{\text{sl}}^l \cdot F \leq (2/\delta)^{n+1-l} Q_{\text{sl}}^{n+1}, \\ K_{F^*} \cdot Q_{\text{sl}}^{l-1} &\leq (K_{Z_{\text{sl}}} + cQ_{\text{sl}})|_F \cdot Q_{\text{sl}}^{l-1}, \quad 0 < c \leq 2(n+1-l)/\delta. \end{aligned}$$

Let W be the image of F in the other copy of X .

Case 4.1. The map $F \rightarrow W$ is generically finite and $W \neq X$. Then $l = \dim W < n$. The two axes restrict effectively, and (7.6), (7.3), and (7.12) give the three bounds (7.4) on F^* , with $Q = Q_{\text{sl}}|_{F^*}$. Theorem 4.3 excludes this case.

Case 4.2. The map $F \rightarrow W$ is not generically finite. Its general fiber is the full projective line, so F is the full bundle over W , by closure of these fibers. Write $w = \dim W = l - 1 < n$. If $w = 0$, its degree is q , and (7.12), (7.7), and (7.8) give

$$q \leq (2/\delta)^n Q_{\text{sl}}^{n+1} \leq (2/\delta)^n (2^{n+1} - 1)qv_+ < q.$$

For $w > 0$, resolve $W^* \rightarrow W$ and use its smooth projective bundle $p: F^* \rightarrow W^*$ as a resolution of F . Let P_W, L_W be pullbacks of P, L , and write $Q = p^*P_W + q\xi$. The divisor $B_W = K_{W^*} + bL_W$, $b = 2n + 1$, is big by Theorem 4.3. Cancelling the identical relative terms $-2\xi + p^*L_W$ in the canonical inequality of (7.12) gives

$$p^*(K_{W^*} - K_X|_{W^*}) \cdot Q^w \leq cQ^{w+1}.$$

Using both nef comparisons yields

$$(7.13) \quad p^*B_W \cdot Q^w \leq \left[c + \frac{1+b/\lambda}{r} \right] Q^{w+1}.$$

All mixed intersections of the big B_W with nef P_W, L_W are nonnegative. In the left side the term with one ξ is $wqB_W \cdot P_W^{w-1}$; all other nonzero terms are nonnegative. Furthermore

$$(7.14) \quad (w+1)qP_W^w \leq Q^{w+1} \leq (2^{w+1} - 1)qP_W^w.$$

The lower inequality and (7.12) bound P_W^w from above. (7.13) and (7.14) bound $B_W \cdot P_W^{w-1}/P_W^w$ from above. Subtract the nonnegative bL_W term to obtain a canonical ratio bound. The map $W^* \rightarrow X$ and pullback of (7.3) now satisfy (7.4), a contradiction. The mixed terms were discarded for the proved-big B_W , not for K_{W^*} .

Case 4.3. The map $F \rightarrow X$ is generically finite and dominant. Then $l = n$ and its degree is an integer $e > 0$. Intersect F with both axes and push to X , obtaining effective integral Weil divisors D_1, D_2 . The quotient of the two coordinate sections is a rational section of the pullback of L or its inverse. Its field norm, after Stein factorization, gives

$$(7.15) \quad D_1 - D_2 \sim \pm eL.$$

The norm retains all mapping and vanishing multiplicities; birational exceptional divisors push to zero. The D_i need not be rational Cartier. If S is either axis, $Q_{\text{sl}} - qS$ and $Q_{\text{sl}} - P$ are nef. Thus

$$qD_i \cdot P^{n-1} \leq qS \cdot F \cdot Q_{\text{sl}}^{n-1} \leq Q_{\text{sl}}^n \cdot F.$$

The right side is bounded by (7.12). Proposition 6.3, applied to the identity cover and these integral cycles, forces $D_1 = D_2 = 0$ for large j . (7.15) would make eL linearly trivial, contradicting strict nefness. This completes the exclusion of all $1 \leq l \leq n$.

Step 5. We reduce to centers dominating both factors with dimension n . Each projection of V has generic fiber dimension either zero or $n+1$. If a projection has dimension $n+1$, V is the full inverse image of a proper subvariety W of that factor, with $\dim W \leq n-1$ because V is proper. Its fibers over the other factor have dimension $\dim W + 1 \in [1, n]$, excluded in **Step 4**. Thus both projections are generically finite onto their images and $f \leq n$. If $f < n$, (7.10), (7.11), and (7.3) give (7.4) for either projection from a resolution, again impossible. Only $f = n$ and dominance over both copies of X remain.

Step 6. We exclude every divisorial branch component of the two Stein covers. Resolve such a V to a smooth projective Y , with $h: Y \rightarrow Z$ and dominant generically finite $g_i: Y \rightarrow X$. Put $Q = h^*R$, $M_i = g_i^*L$. For uniform $B > 0, c \geq 0$,

$$(7.16) \quad 0 < Q^n \leq B, \quad K_Y \cdot Q^{n-1} \leq (c+1/r)Q^n, \quad Q - g_i^*P \text{ nef}, \quad Q - r\lambda M_i \text{ nef}.$$

Their degrees e_i satisfy $e_i v_- \leq Q^n$, so $e_i \leq E$ for one fixed integer $E \geq B/v_-$.

The relative canonical divisor $\text{Ram}_i = K_Y - g_i^*K_X$ is effective even though X may be singular. Take a resolution $w: W \rightarrow X$ with $K_W = w^*K_X + F$, $F \geq 0$, and resolve the rational lift of g_i to obtain $a: Y' \rightarrow Y$ birational and $b: Y' \rightarrow W$ generically finite. The determinant of the differential between smooth varieties gives

$$K_{Y'} - (g_i \circ a)^*K_X = \text{Ram}_b + b^*F \geq 0.$$

Push down by a to obtain $\text{Ram}_i \geq 0$. At a prime mapping onto a divisor of X , its coefficient is the ordinary ramification index minus one, hence an integer at least one when ramified. Fractional coefficients above singular loci do not affect this assertion.

Since $-K_X = tL - D$ and D is nef, (7.16) gives

$$(7.17) \quad \text{Ram}_i \cdot Q^{n-1} \leq \left[c + \frac{1+t/\lambda}{r} \right] Q^n.$$

If the finite normal Stein cover of g_i has a branch prime $J \subset X$, a ramifying prime above it has a strict transform on Y with positive integral coefficient in Ram_i . Birational morphisms to normal varieties are isomorphisms at generic divisorial points. Projection formula and the nef comparison give

$$0 < J \cdot P^{n-1} \leq \text{Ram}_i \cdot (g_i^*P)^{n-1} \leq \text{Ram}_i \cdot Q^{n-1}.$$

This is uniformly bounded by (7.17), contradicting Proposition 6.3 for the identity cover. Thus no branch prime survives for large j .

Step 7. We obtain a finite list of canonical covers over the fixed base. Write $g_i = \tau_i \circ \mu_i$, with $\mu_i: Y \rightarrow T_i$ birational and $\tau_i: T_i \rightarrow X$ finite normal of degree $e_i \leq E$. Purity over the smooth X_{reg} , [Gro03, Exposé X, Theorem 3.1], and **Step 6** make τ_i etale over X_{reg} . It is therefore quasi-etale. Lemma 6.2 gives canonicity and $K_{T_i} = \tau_i^*K_X$, with the canonical index bounded by the fixed index on X . Lemma 7.3 gives a finite list of isomorphism classes of these covers. The open X_{reg} is fixed independently of j .

Step 8. We prove that the projected correspondences are graphs of regular isomorphisms. Let $H_i = \tau_i^*L$, which is strictly nef Cartier, and let $N_Y = h^*\xi$, which is nef Cartier. Pulling back the actual axis sections gives effective integral Cartier divisors

$$(7.18) \quad A_i = h^*S_i = N_Y - \mu_i^*H_i.$$

For every $i, a \in \{1, 2\}$, the cycle $(g_a)_*A_i$ is effective integral Weil, possibly zero. Since $R - qS_i$ and $R - P_a$ are nef,

$$(7.19) \quad q((g_a)_*A_i) \cdot P^{n-1} = qA_i \cdot (g_a^*P)^{n-1} \leq qA_i \cdot Q^{n-1} \leq Q^n \leq B.$$

Proposition 6.3 kills all four cycles for large j . Finiteness of τ_a then makes every component of A_i exceptional for μ_a . In particular A_i is μ_i -exceptional and μ_i -nef, since N_Y is nef and $\mu_i^*H_i$ has degree zero on contracted curves. The negativity lemma [KM98, Lemma 3.39] (see also [Bir07, Lemma 3.0.45]) gives $A_i \leq 0$. Effectivity gives $A_1 = A_2 = 0$, hence

$$(7.20) \quad \mu_1^*H_1 = \mu_2^*H_2 = N_Y$$

as line bundles.

A curve contracted by μ_1 has zero $\mu_2^*H_2$ -degree and is also contracted by μ_2 , since H_2 is strictly nef. If a component of a projective μ_1 -fiber had positive-dimensional μ_2 -image, that image would contain a curve; taking a component above it and general ample cuts would give a curve in the fiber mapping onto it, a contradiction. Thus each component maps to a point. Connectedness of the birational fibers over normal T_1 makes μ_2 constant on the whole fiber.

Let Γ_0 be the reduced image of (μ_1, μ_2) in $T_1 \times T_2$. Its projection to T_1 is proper, birational, and has singleton set-theoretic fibers. It is quasi-finite, hence finite, and normality of T_1 makes it an isomorphism. Interchanging the indices proves that its other projection is also an isomorphism. Thus the projection of V to $X \times X$ is the image of the graph of an actual isomorphism $T_1 \rightarrow T_2$. Strict nefness was used on the finite covers of the original X , not on a small modification.

Step 9. We obtain an open regular orbit on a fixed cover and exclude it. Fix a failure index beyond all the uniform cutoffs. The original moving incidence dominates Z , hence $X \times X$. Its projected points are covered by graphs of isomorphisms among the finite list from **Step 7**. By Theorem 7.4, their isomorphism schemes have countably many finite-type components. If every component had nondominant product evaluation, the closures of their images in the products $T_1 \times T_2$ would be proper. Their images in $X \times X$ would remain proper because the product cover is finite. The dense constructible image of the original incidence contains an open, and an open complex variety cannot be covered by countably many proper closed subsets. Consequently one isomorphism component has product-dominant evaluation.

Fix one isomorphism in it to identify the two covers with T . The component is in a translate of $\text{Aut}^0(T)$. Product-dominance implies that a general point has an n -dimensional orbit. Algebraic group orbits are locally closed, so this orbit is open; since it meets T_{reg} and automorphisms preserve smoothness, it lies there. In characteristic zero the orbit map is separable. Its differential gives n independent global derivations at a point, so T_T is generically generated by global sections.

For the canonical finite cover $\tau: T \rightarrow X$, put $H = \tau^*L$. Proposition 6.8, whose induction premise is only CS_{n-1} , gives

$$K_T + tH = \tau^*(K_X + tL) = \tau^*D \quad \text{ample.}$$

But D is nef and not big, and hence $(\tau^*D)^n = \deg(\tau)D^n = 0$, contradicting ampleness. This excludes the last centers.

Step 10. We verify uniformity and pass from a Seshadri constant to complete jets. All numerical cutoffs depend only on the fixed volume window, n, λ, q, δ , the finitely many lower-dimensional constants in Theorem 4.3, and the degree bounds used in Proposition 6.3. The center dimensions and cover degrees were bounded before they entered the argument. The finite cover list uses the fixed smooth open; its size is irrelevant to the cutoffs. Therefore failures at arbitrarily large j , even with varying prescribed opens, are impossible.

For a point z with $\epsilon(R_j; z) > a > 0$, a rational, blow it up by $\beta: \widehat{Z} \rightarrow Z$ with exceptional E_z . The class $\beta^*R_j - aE_z$ is ample. To see the interior assertion in the blowup characterization [Laz04, Section 5.1], choose $a < a' < \epsilon(R_j; z)$ with $\beta^*R_j - a'E_z$ nef and a small $0 < a_0 < a$

with $\beta^*R_j - a_0E_z$ ample, then express the coefficient a as their strict convex combination. Serre vanishing [Sta26, Lemma 30.17.1, Tag 0B5U], with the fixed twist $-E_z$, gives

$$H^1\left(\widehat{Z}, \mathcal{O}_{\widehat{Z}}(k\beta^*R_j - (ka+1)E_z)\right) = 0$$

for large divisible k . At the smooth center,

$$\beta_*\mathcal{O}_{\widehat{Z}}(-(ka+1)E_z) = \mathfrak{m}_z^{ka+1},$$

and the higher direct images vanish. Locally this follows from the blowup of a regular local ring and the quotients $\mathcal{O}_{\mathbb{P}^{2n}}(j)$, whose positive cohomology vanishes for $j \geq 0$; elsewhere the map is an isomorphism. Leray and projection formula give the corresponding H^1 vanishing on Z , so the ideal restriction sequence gives all jets through ka .

The index of K_X only requires clearing denominators. For each fixed j , k can be increased independently to satisfy the finite kernel and jet conditions. The coefficient q was fixed in **Step 2**; no common k for all j is required. Taking $a = 2n + 2$ at a point with Seshadri constant greater than $2n + 2$ proves the asserted jet conclusion. $\square$

8. A BOUND FOR NEGATIVE COTANGENT SLOPES

In this section, we bound the negative cotangent contribution by the integral degree of covering rational curves, without imposing a sign on the canonical divisor.

Theorem 8.1 (Quantitative Harder–Narasimhan bend-and-break, [She92, Theorem 9.0.2(bis)]). *Let X be a normal projective n -fold and C a sufficiently high geometric generic complete-intersection curve for ample integral polarizations. Let $\mathcal{F}$ be a saturated term in the Harder–Narasimhan filtration of T_X such that $\mu_{\min}(\mathcal{F}|_C) > 0$. For every ample Cartier divisor A there exist rational curves R from a family dominating X with*

$$(8.1) \quad A \cdot R \leq \frac{2nA \cdot C}{c_1(\mathcal{F}) \cdot C}.$$

The precise input is the Harder–Narasimhan form in [She92, Sections 9.1.1–9.1.4], in arbitrary dimension. The positive-slope hypotheses supply the foliation and its characteristic- p closure in that result. For a compatible quantitative foliated statement see [Spi20, arXiv v4, Theorem 2.27]. The constant in (8.1) concerns a chosen positive Harder–Narasimhan term and is independent of the sharper extremal-ray input in Theorem 2.4.

Theorem 8.2 (Harder–Narasimhan restriction, [MR82, Section 6]; [KS06, arXiv v3, Theorem 1.4]). *For a torsion-free sheaf on a smooth complex projective variety with an ample Cartier polarization, sufficiently high general complete-intersection curves carry the restricted Harder–Narasimhan filtration as their Harder–Narasimhan filtration. The curve can be taken smooth and integral and to avoid the non-locally-free loci of all factors and quotients.*

Proposition 8.3. *Let X be smooth projective of dimension $n \geq 2$ and L nef integral Cartier. Assume every integral rational curve meeting the complement of a countable union of proper closed subsets has L -degree at least one. For an ample rational Cartier P , use slopes*

$$(8.2) \quad \mu_P(\mathcal{E}) = \frac{c_1(\mathcal{E}) \cdot P^{n-1}}{\text{rank } \mathcal{E}}.$$

Then $\mu_{\max, P}(T_X) \leq 2nL \cdot P^{n-1}$. If this maximal slope is positive and its first Harder–Narasimhan term has rank d , the sharper bound is $(2n/d)L \cdot P^{n-1}$.

For sufficiently high general complete intersections of classes l_iP , $1 \leq i \leq n-1$, put $\Lambda = \prod_i l_i$. Their smooth curve C satisfies

$$(8.3) \quad [C] = \Lambda P^{n-1}, \quad \mu_{\max}(T_X|_C) = \Lambda \mu_{\max, P}(T_X) \leq 2nL \cdot C, \quad \mu_{\min}(\Omega_X^1|_C) \geq -2nL \cdot C.$$

Curve slopes are ordinary degree divided by rank. If $P - bL$ is nef, $b > 0$, and $v = P^n$, we may take

$$(8.4) \quad \eta = \frac{2n}{b} \Lambda v, \quad \mu_{\min}(\Omega_X^1|_C) \geq -\eta, \quad \frac{\eta}{\Lambda v} = \frac{2n}{b}.$$

In particular $b = ra$ gives normalized loss $2n/(ra)$. For a smooth projective curve, the assertion holds with $C = X$ and constant two.

Proof. We keep the ample polarization defining the filtration fixed and approximate only the independent divisor that measures rational-curve degrees.

Step 1. We fix the filtration and its slope normalization. Choose k such that $H_0 = kP$ is ample Cartier. Slopes for H_0 are k^{n-1} times slopes for P , so their Harder–Narasimhan filtrations coincide. By Theorem 8.2, general complete intersections of $|m_i H_0|$, with the sufficiently-high condition in its restriction sense, preserve this filtration and avoid its non-locally-free loci. With $l_i = m_i k$ and $\Lambda = \prod l_i$, every factor $\mathcal{G}$ satisfies $\mu(\mathcal{G}|_C) = \Lambda \mu_P(\mathcal{G})$. For the numerical argument choose a geometric generic or very general representative; the resulting inequality depends only on its class and therefore holds on every general curve carrying the restricted filtration.

Step 2. We verify the positive-term hypothesis. Put $\mu = \mu_{\max, P}(T_X)$. If $\mu \leq 0$ the conclusion follows from nefness of L . Otherwise let $\mathcal{F}$ be the first term and $d = \text{rank } \mathcal{F}$. Its restriction is semistable of slope $\Lambda \mu > 0$, so $c_1(\mathcal{F}) \cdot C = d \Lambda \mu$. For the integrability check, if $\mathcal{F} \neq T_X$, the quotient has maximal slope less than $\Lambda \mu$, while $\mathcal{F}|_C \otimes \mathcal{F}|_C$ is semistable of slope $2\Lambda \mu$ in characteristic zero. Thus its bracket obstruction to $T_X/\mathcal{F}$ vanishes on these covering curves and hence globally. If $\mathcal{F} = T_X$, it is already integrable. The characteristic- p closure and bounded instability on this fixed curve needed by Theorem 8.1 are part of that input, not an additional assumption here.

Step 3. We preserve the integral degree lower bound under ample approximation. Fix an ample Cartier divisor H . For each positive rational e , $A_e = L + eH$ is ample. Clear its denominator and use Theorem 8.1; division by that denominator gives a covering rational curve R_e with

$$A_e \cdot R_e \leq \frac{2n A_e \cdot C}{c_1(\mathcal{F}) \cdot C}.$$

The covering family has a member meeting the complement of the stipulated countable union, because the base field is $\mathbb{C}$. Thus $L \cdot R_e \geq 1$ and $H \cdot R_e > 0$, giving

$$1 \leq A_e \cdot R_e \leq \frac{2n(L \cdot C + eH \cdot C)}{c_1(\mathcal{F}) \cdot C}.$$

Let $e \rightarrow 0$ through rationals in this scalar inequality. Its fixed quantities give $d\Lambda\mu \leq 2nL \cdot C$, hence the bound $\mu \leq (2n/d)L \cdot P^{n-1}$. No uniform family of curves as e varies is required.

Step 4. We obtain the cotangent and normalized estimates. Duality of curve bundles reverses and negates the Harder–Narasimhan slopes. It gives (8.3). Testing the nef $P - bL$ with P^{n-1} gives $bL \cdot P^{n-1} \leq v$, which gives (8.4) after multiplication by Λ . In dimension one, $\deg T_X = 2 - 2g$. For $g \geq 1$ this is nonpositive; for $g = 0$ the curve itself has integral L -degree at least one. This proves the asserted constant two. Strict nefness is a special case of the hypothesis, with no exceptional closed sets. $\square$

9. A DETERMINANT ESTIMATE WITH NEGATIVE COTANGENT SLOPE

In this section, we prove the Frobenius determinant estimate for fixed complex geometric data, allowing a negative lower cotangent slope.

Set-up 9.1. Let W be smooth connected projective of dimension $n \geq 2$, let D, H be rational Cartier with H ample, and let S be integral Cartier. Fix an integer $q > 0$ and real $r > 1$, $\epsilon > 0$, $\eta \geq 0$. Put $K = K_W$, $v = H^n$, and assume

$$(9.1) \quad v \leq (2\epsilon)^n, \quad KH^{n-1} \leq 2v/r, \quad (2D + qS)H^{n-1} \leq 4v, \quad (14\epsilon)^n < 1/4, \quad 80\epsilon < 3/4.$$

Fix a smooth integral flag $W = W_n \supset W_{n-1} \supset \cdots \supset W_1 = C$, where the successive divisors have integral Cartier classes $h_i = l_i H$, $l_i > 0$. Put $\Lambda = \prod_{i=1}^{n-1} l_i$ and assume $\mu_{\min}(\Omega_W^1|_C) \geq -\eta$ in ordinary curve degree.

On the quotient projective bundle $Z = \mathbb{P}_{W \times W}(\mathcal{O}_W(S)_1 \oplus \mathcal{O}_W(S)_2)$, let $\xi = c_1(\mathcal{O}_Z(1))$ and $M = D_1 + D_2 + q\xi$. Fix a positive integer k_0 with $k_0 D$ Cartier and finitely many sections of $k_0 M$ generating jets through order $(2n+2)k_0$ at a point z_* off the axes. Fix also $x \in W$, the blowup $\pi: \widehat{W} \rightarrow W$ at x with exceptional J , and

$$(9.2) \quad \gamma = f_*(A_1 \cdots A_{n-1}), \quad J \cdot \gamma > 0,$$

where $f: Y \rightarrow \widehat{W}$ is a smooth projective birational model and all A_i are ample integral Cartier. Assume

$$(9.3) \quad \pi^*(D + K/2 + \lambda S) \cdot \gamma \leq 40\epsilon J \cdot \gamma \quad (0 \leq \lambda \leq q).$$

The two endpoint inequalities suffice, by affinity. All these data are fixed over $\mathbb{C}$.

Theorem 9.2 (Fixed-data determinant estimate). *In Set-up 9.1, put*

$$(9.4) \quad \delta = \frac{\eta}{\Lambda v}, \quad \Theta_\eta = v(7 + n\delta)^n.$$

Choose an ample integral T_0 such that every $T_0 + aS$, $0 \leq a \leq (k_0 - 1)q$ integral, has sections nonzero at the two base coordinates of z_* , and fix such sections. Spread the data over a finitely generated integral $\mathbb{Z}$ -algebra. In every sufficiently large residual characteristic p , after restricting the base to an open, let $F: W \rightarrow W'$ be relative Frobenius, and let x' and S' denote the Frobenius field twists of x and S on W' , respectively. Define

$$(9.5) \quad \begin{aligned} N_p &= \lfloor p/k_0 \rfloor, & B_p &= k_0 N_p D + T_0, & s &= p^n, \\ \mathcal{E}_p &= F_* \mathcal{O}_W(B_p), & U_a &= H^0(W, B_p + paS), & R &= s^2. \end{aligned}$$

The evaluation

$$(9.6) \quad \bigoplus_{\substack{a+b=q \\ a, b \geq 0}} U_a \otimes U_b \otimes \mathcal{O}_{W' \times W'}(-aS'_1 - bS'_2) \rightarrow \mathcal{E}_p \boxtimes \mathcal{E}_p$$

has generic rank R . Fix bases of the source spaces. Any R generically independent columns have a nonzero determinant σ in $\mathcal{Q}_1 \boxtimes \mathcal{Q}_2$, where

$$(9.7) \quad \mathcal{Q}_i = (\det \mathcal{E}_p)^{\otimes s} \otimes \mathcal{O}_{W'}(m_i S'),$$

and m_i is the sum of the selected weights in slot i . For constants $C_1, C_\gamma \geq 0$ depending only on the fixed data,

$$(9.8) \quad 1 - \Theta_\eta - \frac{C_1}{p} \leq \frac{\text{ord}_{(x', x')} \sigma}{R} \leq 80\epsilon + \frac{2C_\gamma}{p}.$$

The constants are uniform in p , filtration degree, and the chosen columns. Consequently these data cannot coexist if

$$(9.9) \quad 80\epsilon + v \left(7 + \frac{n\eta}{\Lambda v} \right)^n < 1.$$

All assertions remain valid with 7 replaced by $4 + 2/r$, after recomputing the constants. If $E_\eta = v((7 + n\delta)^n - 7^n)$, the lower bound is also $3/4 - E_\eta - C_1/p$. Thus $80\epsilon + E_\eta < 3/4$ is another sufficient incompatibility condition; the upper bound has no additional η term.

Proof. We adapt the finite-data construction of [Ope26b, Theorem 6.1 and Section 6.6], retaining the loss from negative cotangent slope. We spread a finite collection of complex data before considering any residual characteristic.

Step 1. We preserve the fixed flag, its slope bound, and the numerical data. The flag satisfies

$$(9.10) \quad [C] = \Lambda H^{n-1}, \quad H \cdot C = \Lambda v, \quad d_i = h_i \cdot C = l_i \Lambda v > 0.$$

The d_i are integers. Choose T_0 and the stipulated nonvanishing sections sufficiently positive, and spread the varieties, divisors, maps, flag, points, and finite jet system over a finitely generated integral $\mathbb{Z}$ -subalgebra of $\mathbb{C}$. Shrink so all varieties and flag members remain smooth projective geometrically integral, ample divisors remain ample, the chosen sections remain nonzero at the specified points, and the jet surjection remains surjective. Spread fixed isomorphisms on dense opens to preserve birationality. Intersection numbers and the genus g of C stay constant; exclude primes dividing the rational denominators.

We justify preservation of $\mu_{\min}(\Omega_W^1|_C) \geq -\eta$, also for real η . For the relative curve bundle $\mathcal{V}$ of rank n , choose a relative ample twist of fixed integral degree $d \geq 0$ making it globally generated. Every rank- a coherent quotient has degree at least $-ad$, because its torsion-free part after twisting is generated and its torsion adds nonnegative degree. A quotient of slope below $-\eta$ thus has $1 \leq a \leq n$ and integer degree in the finite interval $[-ad, -a\eta]$. Each such Hilbert polynomial gives a proper relative Quot scheme. Its geometric generic fiber is empty: removing torsion from a hypothetical quotient only lowers its degree, contrary to the complex minimum-slope bound. Delete the finitely many closed images of these schemes. The lower slope bound then holds on every remaining geometric fiber.

Such a characteristic-zero integral finite-type base has closed points of arbitrarily large residual characteristic: its dominant constructible image in $\text{Spec } \mathbb{Z}$ contains an open. Work over algebraic closures of those residue fields. The fixed class γ still pairs nonnegatively with every effective divisor on the reduced blowup, since its pullback to the fixed Y is effective and has nonnegative intersection with the ample A_i . This applies to new effective divisors on a reduction, without requiring them to lift.

Step 2. We obtain generic full rank from ordinary jets. Set $a_p = p - k_0 N_p \in \{0, \dots, k_0 - 1\}$, so $B_p = pD + T_0 - a_p D$. Products of N_p members of the fixed jet system generate jets through order $(2n+2)k_0 N_p$. To check this without dividing by factorials, choose sections with prescribed leading monomials through the original order. Every monomial of degree at most the enlarged order is a product of N_p monomials within the original bound. Their products have the desired leading terms; triangular elimination by degree spans the truncated local ring.

Let $d_p = a_p q$. A fixed section of T_0 in the first slot and of $T_0 + d_p S$ in the second, multiplied by the second fiber coordinate to exponent d_p , gives a section of $d_p \xi + T_{0,1} + T_{0,2}$ nonzero at z_* . Multiplication preserves the jet span, so sections of $pq\xi + B_{p,1} + B_{p,2}$ generate those jets. The quotient by the p th powers of regular parameters at the $(2n+1)$ -dimensional smooth point z_* has largest monomial degree $(2n+1)(p-1)$, bounded by $(2n+2)k_0 N_p$ for large p . The sections therefore surject onto this quotient.

Relative Frobenius of smooth W is finite flat of rank $s = p^n$, as seen from the local monomial basis with exponents less than p . Projection formula gives (9.6), since $F^* S' = pS$. Near the base coordinates of z_* trivialize both summands and let z be the fiber coordinate, with value $z_0 \neq 0$. If A_* is the tensor product of the two base Frobenius-fiber algebras, then $\dim A_* = p^{2n} = R$ and the total quotient is

$$A_*[z]/(z^p - z_0^p),$$

free over A_* on $1, z, \dots, z^{p-1}$. A section of fiber weight j has coefficient in

$$H^0(W, B_p + jS) \otimes H^0(W, B_p + (pq - j)S).$$

Projection of the surjective jet map to the coefficient of 1 retains exactly $j = pa$, with nonzero factor z_0^{pa} . Their coefficients are the values of (9.6) and span A_* . Hence the evaluation has rank R at that base pair, and therefore generically. The algebra A_* is allowed to be nonreduced.

Step 3. We bound all graded pieces of the Frobenius diagonal. Put $\Delta_F = W \times_{W'} W$, and let $h = F \circ \text{pr}_1 = F \circ \text{pr}_2$. Its reduction is the ordinary diagonal. On it, $pS_1 = pS_2 = h^* S'$, so every weight pair $a + b = q$ restricts to the common line bundle

$$\mathcal{L}_p = \mathcal{O}_{\Delta_F}(B_{p,1} + B_{p,2} + pqS_1).$$

Finite flat base change and projection formula identify $h_*\mathcal{L}_p$ with $(\mathcal{E}_p \otimes \mathcal{E}_p)(qS')$. Thus the evaluation rank at a diagonal point is at most $h^0(\Delta_F, \mathcal{L}_p)$.

For a rank- n bundle V in characteristic p , let $A_j(V)$ be the degree- j part of the symmetric algebra modulo the p th powers of its degree-one generators. Additivity of p th powers makes the definition independent of frame. Put $u = n(p-1)$ and $e_j = \text{rank } A_j(V)$. The monomial basis gives

$$(9.11) \quad e_j = e_{u-j}, \quad \sum_{j=0}^u e_j = p^n = s.$$

These bundles are locally free and quotients of $V^{\otimes j}$. In smooth coordinates, the ideal of the reduced diagonal is generated by coordinate differences whose p th powers vanish, and its degree-one conormal is Ω_W^1 . Its ordinary ideal filtration therefore has graded pieces

$$(9.12) \quad \mathcal{G}_j = \mathcal{O}_W(2B_p + pqS) \otimes A_j(\Omega_W^1), \quad 0 \leq j \leq u.$$

Using only the second-factor bundle instead gives graded pieces $\mathcal{O}_W(B_p) \otimes A_j(\Omega_W^1)$ for $F^*\mathcal{E}_p$.

Multiplication into the top truncated degree is a perfect pairing, since each exponent vector pairs with its complement to $(p-1, \dots, p-1)$. The top line has character $(\det V)^{p-1}$, computed on diagonal matrices and hence on GL_n . Consequently

$$(9.13) \quad A_{u-j}(V) = A_j(V)^\vee \otimes (\det V)^{p-1}.$$

We use [Lan04a, Corollary 2.5], unchanged by [Lan04b], in its curve form. If A_C is nef and $T_C \otimes A_C$ is globally generated, a rank- a bundle E satisfies

$$(9.14) \quad \mu_{\min}(E) - L_{\min}(E) \leq \frac{(a-1)\deg A_C}{p-1}, \quad L_{\min}(E) = \lim_{e \rightarrow \infty} p^{-e} \mu_{\min}(F_C^{e*}E).$$

For $g \leq 1$ take $A_C = \mathcal{O}_C$. For $g \geq 2$, a line bundle of degree $4g-2$ makes $T_C \otimes A_C$ have degree $2g$, hence globally generated by Serre duality [Sta26, Remark 53.4.3, Tag 0BS4] and point evaluation. With

$$d_g = 0 \ (g \leq 1), \quad d_g = 4g-2 \ (g \geq 2), \quad c = (n-1)d_g,$$

we obtain, for $V = \Omega_W^1|_C$,

$$(9.15) \quad L_{\min}(V) \geq -\eta - c/(p-1).$$

For every integer $i \geq 0$ we also have $\mu_{\min}(V^{\otimes i}) \geq iL_{\min}(V)$. Here is the needed tensor argument in positive characteristic. Fix p, i and choose A_e of degree $\lceil -\mu_{\min}(F_C^{e*}V) \rceil + 2g$. The bundle $F_C^{e*}V \otimes A_e$ is generated: after subtracting any point its minimum slope is at least $2g-1 > 2g-2$, so Serre duality [Sta26, Remark 53.4.3, Tag 0BS4] gives the required H^1 vanishing. For a vector-bundle quotient Q of $V^{\otimes i}$, $F_C^{e*}Q \otimes A_e^{\otimes i}$ is generated. Its determinant has nonnegative degree, giving $p^e \mu(Q) + i \deg A_e \geq 0$. Division by p^e and the limit $e \rightarrow \infty$ prove the bound. This limit is taken for each fixed p, i , and assumes no tensor semistability in characteristic p .

Since $A_{u-j}(V)$ is a quotient of $V^{\otimes(u-j)}$, (9.13) and (9.15) give

$$(9.16) \quad \begin{aligned} \mu_{\max}(A_j(V)) &= (p-1)K \cdot C - \mu_{\min}(A_{u-j}(V)) \\ &\leq (p-1)K \cdot C + (u-j)\eta + \frac{(u-j)c}{p-1} \\ &\leq (p-1)K \cdot C + (u-j)\eta + nc. \end{aligned}$$

The maximal extra loss is $n(p-1)\eta$.

Step 4. We bound all global sections by one flag polynomial. Define the fixed nonnegative constant

$$c_0 = nc + \max_{0 \leq a < k_0} |(2T_0 - 2aD - K) \cdot C|.$$

(9.1), (9.10), and (9.16) give, uniformly in j ,

$$\begin{aligned} \mu_{\max}(\mathcal{G}_j|_C) &\leq p(2D + qS + K) \cdot C + (u - j)\eta + c_0 \\ &\leq p((4 + 2/r)\Lambda v + n\eta) + c_0 \\ (9.17) \quad &\leq M_p := p(7\Lambda v + n\eta) + c_0. \end{aligned}$$

Successive divisor restriction sequences down the fixed flag yield

$$(9.18) \quad h^0(W, \mathcal{G}_j) \leq \sum_{b_1, \dots, b_{n-1} \geq 0} h^0 \left(C, \mathcal{G}_j|_C \left(- \sum_i b_i h_i|_C \right) \right).$$

For each fixed p, j and each earlier twist, sufficiently negative twisting by the next ample divisor kills the remainder. Enlarging the resulting finite nonnegative sums to (9.18) needs no uniform stopping index. A summand vanishes if $\sum b_i d_i > M_p$, since its maximal slope is negative. Each other summand is at most $e_j(M_p + 1)$: evaluation at $\lfloor M_p \rfloor + 1$ distinct points is injective because its kernel has negative maximal slope. At most $\prod_i (1 + M_p/d_i)$ indices contribute. Thus

$$(9.19) \quad h^0(W, \mathcal{G}_j) \leq e_j P_\eta(p), \quad P_\eta(p) = (M_p + 1) \prod_{i=1}^{n-1} (1 + M_p/d_i).$$

This single polynomial has degree n , nonnegative coefficients, and leading coefficient

$$\frac{(7\Lambda v + n\eta)^n}{\prod_i d_i} = \frac{(7\Lambda v + n\eta)^n}{\Lambda^n v^{n-1}} = \Theta_\eta.$$

Put $C_1 = P_\eta(1) - \Theta_\eta \geq 0$. Dividing the lower-degree terms by p^n bounds them by C_1/p for $p \geq 1$. Summing the filtration and using (9.11) gives

$$(9.20) \quad h^0(\Delta_F, \mathcal{L}_p) \leq s P_\eta(p) \leq R(\Theta_\eta + C_1/p).$$

The sum of the ranks is s ; no extra factor counting the filtration pieces appears.

Step 5. We compute the determinant line bundles and bound their vanishing orders. Choose any R independent columns in (9.6). Their slot weights satisfy $0 \leq m_i \leq qR$, so $\lambda_i = m_i/R \in [0, q]$. The source determinant is $\mathcal{O}(-m_1 S') \boxtimes \mathcal{O}(-m_2 S')$, while the target determinant is $(\det \mathcal{E}_p)^s \boxtimes (\det \mathcal{E}_p)^s$. Thus the determinant is a nonzero section of the actual line bundles in (9.7).

Pairing indices in (9.13) gives

$$\sum_{j=0}^u c_1(A_j(\Omega_W^1)) = \frac{s(p-1)}{2} K.$$

For each pair the contribution is $e_j(p-1)K$; summing that identity over every j and dividing by two also covers a possible middle degree. The filtration of $F^* \mathcal{E}_p$ therefore gives $c_1(F^* \mathcal{E}_p) = sB_p + s(p-1)K/2$. Under numerical identification with the field twist, Frobenius pullback multiplies divisor classes by p . Hence

$$(9.21) \quad \frac{c_1(\mathcal{Q}_i)}{R} = D + K/2 + \lambda_i S + \frac{T_0 - a_p D - K/2}{p}.$$

No cotangent slope enters this identity.

Set

$$C_\gamma = \max_{0 \leq a < k_0} \frac{|\pi^*(T_0 - aD - K/2) \cdot \gamma|}{J \cdot \gamma}.$$

The fixed interval inequality (9.3) implies

$$\frac{\pi^* c_1(\mathcal{Q}_i) \cdot \gamma}{RJ \cdot \gamma} \leq 40\epsilon + C_\gamma/p.$$

For any nonzero section τ_i of $\mathcal{Q}_i$ on the reduction, the strict transform of its effective divisor has nonnegative intersection with the twist of γ . Its class is $\pi^*c_1(\mathcal{Q}_i) - \text{ord}_{x'}(\tau_i)J$. Consequently

$$(9.22) \quad \text{ord}_{x'}(\tau_i) \leq R(40\epsilon + C_\gamma/p).$$

This is uniform in the chosen columns and does not require lifting τ_i .

Degree-zero Künneth identifies the section space of $\mathcal{Q}_1 \boxtimes \mathcal{Q}_2$ with the tensor product of the two section spaces. Choose bases adapted to the decreasing filtrations by order at x' . Within each order their initial forms are independent; tensor products are independent in each bidegree in the disjoint local parameter sets. Distinct bidegrees are distinct direct summands of the total graded local ring. Thus a nonzero tensor has order at most the sum of the maximal factor orders. (9.22) gives

$$(9.23) \quad \text{ord}_{(x',x')} \sigma \leq R(80\epsilon + 2C_\gamma/p).$$

Step 6. We compare the diagonal rank defect with the upper order. At (x',x') , (9.20) bounds the matrix rank by $R(\Theta_\eta + C_1/p)$. If its actual rank is t , invertible constant row operations put $R - t$ rows in the maximal ideal. Every determinant term contains one entry from each such row and has order at least $R - t$. The determinant changes only by a nonzero scalar. Therefore

$$\text{ord}_{(x',x')} \sigma \geq R(1 - \Theta_\eta - C_1/p),$$

which with (9.23) proves (9.8). If $\Delta = 1 - 80\epsilon - \Theta_\eta > 0$, take an available residual characteristic exceeding $(C_1 + 2C_\gamma)/\Delta$ and all fixed thresholds. This contradicts (9.8) and proves (9.9).

Finally $\Theta_0 = 7^n v \leq (14\epsilon)^n < 1/4$ and $\Theta_\eta = \Theta_0 + E_\eta$, which gives the stated $3/4$ form and sufficient conditions. The middle line of (9.17) proves the entire argument with $4 + 2/r$ replacing 7. The loss is $\eta/(\Lambda v)$: increasing flag degrees multiplies both η and Λ when slopes come from an ambient restriction theorem, so it does not by itself reduce the normalized loss. $\square$

10. A COMMON MOVABLE WITNESS

In this section, we assemble the scalar and jet data on a resolution and obtain one movable class controlling both determinant factors.

Theorem 10.1 ([BDPP13, author final, Theorem 2.2 and Definition 1.3(v)]). *On a smooth complex projective variety, an algebraic real divisor class is pseudo-effective if and only if it has nonnegative intersection with every class $f_*(A_1 \cdots A_{n-1})$, where f is a smooth projective birational model and the A_i are ample integral Cartier divisors. Such classes generate the strongly movable cone.*

We use the algebraic divisor-class assertion; no transcendental extension is needed.

Proposition 10.2 (Assembly on the original variety). *Let X be normal projective of dimension $n \geq 2$, with K_X rational Cartier, and let L be nef integral Cartier. Assume every rational integral curve meeting the complement of a countable union of proper closed subsets has L -degree at least one. Let P be ample rational Cartier, let $r > 1$, $b > 0$, $\epsilon > 0$ be real, and let $q > 0$ be integral. Suppose*

$$(10.1) \quad P - bL \text{ nef}, \quad P/r - K_X \text{ nef}, \quad b > 2q,$$

and, writing $v_0 = P^n$, $\rho = v_0^{1/n}$, suppose

$$(10.2) \quad 0 < v_0 < (2\epsilon)^n, \quad (14\epsilon)^n < 1/4, \quad 80\epsilon < 3/4.$$

Assume a smooth point $y \in X$ satisfies $\text{ord}_y(s) \leq 4kp$ for every nonzero section of every Cartier kP . On

$$(10.3) \quad Z_X = \mathbb{P}_{X \times X}(\mathcal{O}_X(L)_1 \oplus \mathcal{O}_X(L)_2), \quad M_X = P_1 + P_2 + q\xi_X,$$

assume some finite complete system in degree k_0 , with k_0P Cartier, generates jets through order $(2n+2)k_0$ at a torus point over two smooth points of X . Then

$$(10.4) \quad 80\epsilon + v_0 \left(7 + \frac{2n^2}{b}\right)^n \geq 1.$$

It suffices instead to assume a Seshadri constant greater than $2n+2$ at such a point. No sign assumption on K_X or singularity assumption beyond the stated normality and rational Cartierness is required.

Proof. Choose a resolution $w: W \rightarrow X$ isomorphic over X_{reg} and put $D = w^*P$, $S = w^*L$. We construct the fixed data of Set-up 9.1, including a common witness for its full interval of coefficients.

Step 1. We obtain a compatible ample flag with controlled cotangent loss. Write $K_W = w^*K_X + E_+ - E_-$, where E_+, E_- are effective exceptional rational divisors without common components. Projection formula gives

$$K_W D^{n-1} = K_X P^{n-1} \leq v_0/r < 2v_0/r, \quad (2D + qS)D^{n-1} \leq (2 + q/b)v_0 < 4v_0.$$

Fix ample integral A_W . For every sufficiently small positive rational τ , $H = D + \tau A_W$ is ample, $H - bS$ is nef, and continuity preserves

$$(10.5) \quad v = H^n \leq (2\epsilon)^n, \quad K_W H^{n-1} \leq 2v/r, \quad (2D + qS)H^{n-1} \leq 4v.$$

On W , exclude the exceptional locus and inverse images of the given countable closed sets. A rational curve meeting the remaining isomorphism open cannot be contracted. It maps to an integral rational curve in the allowed part of X , and its S -degree is the positive mapping degree times that curve's L -degree, hence at least one. Proposition 8.3 applies to the nef Cartier S , though it need not be strictly nef. Choose successively sufficiently high general divisors in $|l_i H|$ in the restriction-theorem range. Their smooth flag ends in C , and

$$(10.6) \quad [C] = \Lambda H^{n-1}, \quad \mu_{\min}(\Omega_W^1|_C) \geq -\eta, \quad \eta = \frac{2n}{b}\Lambda v.$$

The flag is chosen over $\mathbb{C}$.

Step 2. We pull back the finite jet system. Base change gives $Z_W = \mathbb{P}_{W \times W}(\mathcal{O}_W(S)_1 \oplus \mathcal{O}_W(S)_2) \rightarrow Z_X$. It is an isomorphism near the chosen jet point since both base coordinates lie in X_{reg} . Pullback sends M_X to $D_1 + D_2 + q\xi_W$ and preserves its jet surjection. A finite basis supplies the required finite system. For the Seshadri alternative, the tautological class ξ_X is nef because both defining summands are nef line bundles, and it is relatively ample. The class $P_1 + P_2$ is ample on the base. For a sufficiently large integer $m > 1/q$, the class $\xi_X + m(P_1 + P_2)$ is ample, and

$$M_X = \frac{1}{m}(\xi_X + m(P_1 + P_2)) + \left(q - \frac{1}{m}\right)\xi_X.$$

Thus M_X is ample as the sum of an ample rational class and a nef class. Choose rational a strictly between $2n+2$ and that Seshadri constant. The blowup at the smooth point and Serre vanishing [Sta26, Lemma 30.17.1, Tag 0B5U] with twist $-E$ give jets through $k_0 a$ for every sufficiently large divisible k_0 , by the local blowup ideal identity and its higher-direct-image vanishing. Thus these include the required order.

Step 3. We build a scalar envelope that absorbs both discrepancy signs. Let $x = w^{-1}(y)$ and put $D^+ = 2D + E_+$, which is big. For divisible m , normality gives $w_*\mathcal{O}_W(mE_+) = \mathcal{O}_X$: a rational function allowed exceptional poles has none at the strict transform of any prime of X , so it extends across the codimension-two complement. Projection formula identifies sections of mD^+ with sections of $2mP$, multiplied by the canonical exceptional section. Since x is off E_+ ,

$$(10.7) \quad \text{ord}_x(s) \leq 8m\rho < 16m\epsilon \quad (0 \neq s \in H^0(W, mD^+)).$$

Blow up x by $\pi: \widehat{W} \rightarrow W$ with exceptional J . The class $\pi^*D^+ - 40\epsilon J$ is not pseudo-effective. Otherwise its convex combination with the big π^*D^+ would make $\pi^*D^+ - cJ$ big for any rational $8\rho < c < 40\epsilon$. A divisible multiple would have a nonzero section, descending to a section of mD^+ of order at least $mc > 8m\rho$, contradicting (10.7). The descent uses $\pi_*\mathcal{O}(-mcJ) = \mathfrak{m}_x^{mc}$.

Step 4. We select one movable class for the whole interval. By Theorem 10.1, there exists a single class $\gamma = f_*(A_1 \cdots A_{n-1})$ with

$$(10.8) \quad \pi^*D^+ \cdot \gamma < 40\epsilon J \cdot \gamma.$$

If separation first gives a positive combination or limit of generators, strict negativity selects one generator with negative pairing. Since D^+ is big, its pairing is nonnegative, so $J \cdot \gamma > 0$. For $0 \leq \lambda \leq q$, the identity

$$(10.9) \quad P - \lambda L - K_X/2 = \left(1 - \frac{\lambda}{b} - \frac{1}{2r}\right)P + \frac{\lambda}{b}(P - bL) + \frac{1}{2}(P/r - K_X)$$

is nef, with positive first coefficient by $b > 2q$ and $r > 1$. On W ,

$$(10.10) \quad D^+ - (D + K_W/2 + \lambda S) = w^*(P - \lambda L - K_X/2) + \frac{1}{2}(E_+ + E_-)$$

is pseudo-effective. Pullback pairs nonnegatively with γ , so (10.8) proves

$$\pi^*(D + K_W/2 + \lambda S) \cdot \gamma < 40\epsilon J \cdot \gamma \quad (0 \leq \lambda \leq q).$$

Both endpoints use this one fixed class; no sign of K_W or of its exceptional discrepancy was assumed.

Step 5. We choose the perturbation before reduction and apply the determinant estimate. If (10.4) failed, choose the rational $\tau > 0$ in **Step 1** small enough, using strictness and $H^n \rightarrow P^n$, that

$$80\epsilon + v(7 + 2n^2/b)^n < 1.$$

Fix this τ , the flag, points, finite jet system, and witness over $\mathbb{C}$. (10.5) and (10.6) and **Steps 2–4** verify Set-up 9.1. Its normalized loss is $\eta/(\Lambda v) = 2n/b$. Theorem 9.2 contradicts the displayed inequality. This proves (10.4).

For an arbitrary lower slope bound $-\eta$ on this flag, the precise sufficient budget is

$$(10.11) \quad \frac{\eta}{\Lambda v} < \frac{((1 - 80\epsilon)/v)^{1/n} - 7}{n}.$$

The right side is positive because $7^n v \leq (14\epsilon)^n < 1/4 < 1 - 80\epsilon$. All choices were fixed before residual characteristic grew; no perturbation, flag degree, or index limit is taken afterwards. $\square$

11. CANONICAL INDUCTION AND SHARP AMPLENESS

In this section, we complete the induction on the entire canonical category and prove Theorems B and A.

Proof of Theorem B. We prove CS_n by strong induction, with the specific big divisor $K_X + (2n + 1)L$, and then use Lemma 2.5. The inductive category consists of all normal projective canonical varieties, without $\mathbb{Q}$ -factoriality or a bound on the canonical index.

Step 1. We establish the one-dimensional case. A normal complex projective curve is smooth. If its genus is g and $\deg L = \ell \geq 1$, then

$$\deg(K_X + 3L) = 2g - 2 + 3\ell \geq 1.$$

Positive degree is equivalent to both ampleness and bigness on a projective curve, giving CS_1 . Also $\deg(K_X + tL) \geq t - 2 > 0$ for real $t > 2$, which is the claimed sharp range.

Step 2. We normalize a hypothetical nonbig adjoint in dimension n . Let $n \geq 2$, assume CS_f for every $1 \leq f < n$, and take arbitrary canonical X and strictly nef Cartier L in dimension n . Put $t_0 = 2n + 1$, $D = K_X + t_0L$, and $N = K_X + 2nL$. Proposition 2.3 makes N nef. Suppose

$D = N + L$ is not big. Fix ample Cartier A and $\epsilon = 1/1000$. Lemma 2.6 gives $P_j = r_j(D + u_j A)$ with

$$(11.1) \quad r_j \rightarrow \infty, \quad v_j = P_j^n \rightarrow \epsilon^n, \quad P_j - r_j L \text{ nef}, \quad P_j/r_j - K_X = t_0 L + u_j A \text{ nef}.$$

Every actual integral curve has P_j -degree at least r_j .

Step 3. We fix the two-slot coefficient and then one large normalization index. Theorem 7.5, under $\text{CS}_{<n}$, supplies a single integer $q > 0$ independent of j , taking $h = 2n + 2$. For every sufficiently large j , it supplies a smooth torus point over $X_{\text{reg}} \times X_{\text{reg}}$ with the required finite complete jet system for

$$Z_X = \mathbb{P}_{X \times X}(\mathcal{O}_X(L)_1 \oplus \mathcal{O}_X(L)_2), \quad P_{j,1} + P_{j,2} + q\xi_X.$$

Theorem 5.2 applies to $(X, 0)$, with $\lambda = t_0 - 2n = 1$ and window $\epsilon^n/2 \leq v_j \leq 2\epsilon^n$. It gives a smooth y_j such that every nonzero section of every Cartier kP_j satisfies $\text{ord}_{y_j}(s) \leq 4kv_j^{1/n}$. The very-general complement is nonempty over $\mathbb{C}$. The point y_j may differ from both base coordinates of the jet point; Proposition 10.2 allows these independent choices.

Now, after fixing q , choose one j satisfying all eventual conditions and

$$(11.2) \quad r_j > \max\{1, 2q, 2n^2\}, \quad 0 < v_j < (2\epsilon)^n.$$

Such a j exists by (11.1). The chosen ϵ satisfies $(14\epsilon)^n < 1/4$ and $80\epsilon < 3/4$ for every $n \geq 2$.

Step 4. We obtain the determinant contradiction on the original X . Use Proposition 10.2 with $P = P_j$, $r = b = r_j$, and the fixed q, ϵ . Normality and rational Cartierness of K_X follow from canonicity. Strict nefness and integrality of L give degree at least one on every integral rational curve, so the exceptional countable set may be empty. All nef, volume, scalar, and jet hypotheses are verified by (11.1), (11.2), and **Step 3**. The necessary inequality is therefore

$$(11.3) \quad \begin{aligned} 1 &\leq 80\epsilon + v_j(7 + 2n^2/r_j)^n \\ &< 80\epsilon + (16\epsilon)^n \leq \frac{80}{1000} + \left(\frac{16}{1000}\right)^2 < 1. \end{aligned}$$

For the strict inequality, $r_j > 2n^2$ and $v_j < (2\epsilon)^n$ give $7 + 2n^2/r_j < 8$. The final bound uses $n \geq 2$ and $16/1000 < 1$. This contradiction proves $D = K_X + (2n + 1)L$ big.

The construction uses a resolution only inside Proposition 10.2. There w^*L is merely nef; the rational-curve degree bound holds on curves meeting the isomorphism open. The ample flag class is $w^*P_j + \tau A_W$, with τ chosen after j . Its normalized slope loss is $2n/r_j$. Exceptional canonical intersections vanish at the unperturbed class, strict margins permit that perturbation, and $2w^*P_j + E_+$ supplies the common scalar envelope. Thus no strict positivity on exceptional curves and no pseudo-effectivity of K_X or of the resolution's canonical divisor was introduced.

Step 5. We close the canonical induction and recover the sharp threshold. Since X, L in dimension n were arbitrary, we have proved CS_n for the full canonical category, with the positive rational coefficient $2n + 1$. The induction for dimension $n + 1$ therefore includes every canonical n -fold that can occur as a modification or cover. Lemma 2.5, applied to the rational klt pair $(X, 0)$, now turns the big adjoint at $2n + 1$ into ampleness for every real $t > n + 1$.

The order of choices throughout is dimension and X, L ; t_0, ϵ ; the normalized sequence; fixed q ; one sufficiently large j ; scalar and jet points and one finite divisible jet degree; a resolving perturbation, flag, and one movable witness over $\mathbb{C}$; finally residual characteristic. No parameter is taken to a limit together with the residual characteristic. Every induction use in dimension n involved only CS_f with $f < n$. $\square$

Proof of Theorem A. A smooth variety is canonical, so Theorem B applies. Equivalently, the induction proves $K_X + (2n + 1)L$ big on smooth X , and the smooth setting of Proposition 2.3 recovers ampleness for every real $t > n + 1$ directly from the smooth cone bound. For a curve, $\deg(K_X + tL) \geq t - 2 > 0$ gives the conclusion directly. $\square$

12. TRANSFER TO KLT PAIRS AND REAL BOUNDARIES

In this section, we prove Corollary 1.1 by transferring canonical bigness in a fixed dimension to rational pairs and then preserving the Cartier condition in a finite rational convex decomposition.

Proposition 12.1 (Category transfer in a fixed dimension). *For $n \geq 1$, CS_n is equivalent to ampleness of $K_X + B + tL$ for every real $t > 2n$, every projective klt rational n -fold pair (X, B) , and every strictly nef Cartier L .*

More generally, under CS_n , let X be any normal integral projective n -fold, let $B \geq 0$ be a rational Weil divisor with $R = K_X + B$ rational Cartier, and let L be strictly nef Cartier. Then $R + tL$ is big for every real $t > 2n$, without a singularity condition on the pair or an upper bound on its boundary coefficients. No separate $\mathbb{Q}$ -Cartierness of K_X is required.

Proof. Assume CS_n . Take a smooth projective resolution $\pi: W \rightarrow X$ and put $M = \pi^*L$. Theorem 4.3, now in dimension $f = n$, gives $A_q = K_W + qM$ big for every rational $q > 2n$. This use is justified by the premise in dimension n , and does not assert strict nefness of M .

Choose compatible canonical divisors. On every nonexceptional prime of W , $K_W - \pi^*R$ has coefficient minus the corresponding coefficient of B . If its exceptional part is $\sum a_i E_i$, define $E = \sum \max\{a_i, 0\} E_i$. Then $E \geq 0$ is exceptional and

$$(12.1) \quad K_W - \pi^*R \leq E, \quad A_q \leq \pi^*(R + qL) + E.$$

For divisible k , normality gives $\pi_*\mathcal{O}_W(kE) = \mathcal{O}_X$. To check this, a rational function allowed poles only at exceptional primes has no pole at a codimension-one point of X and is regular; the reverse inclusion follows from effectivity. Projection formula therefore gives

$$(12.2) \quad h^0(W, kA_q) \leq h^0(W, k\pi^*(R + qL) + kE) = h^0(X, k(R + qL)).$$

The left side has positive order- k^n growth for large divisible k . Hence $R + qL$ is big. For real $t > 2n$, choose rational $2n < q < t$ and add the nef $(t - q)L$, preserving the interior of the pseudo-effective cone.

For a klt rational pair, Proposition 2.3 turns this bigness into ampleness for every $t > 2n$. Conversely, apply the asserted rational-pair ampleness to $(V, 0)$ for canonical V , and take $t = 2n + 1$ to obtain CS_n . Thus the equivalence and the more general bigness statement hold. $\square$

Lemma 12.2 (Rational boundaries with a fixed Cartier presentation). *Let X be normal projective and $B \geq 0$ a real Weil divisor of finite support with $K_X + B$ real Cartier. If (X, B) is klt, respectively lc, then*

$$(12.3) \quad B = \sum_{a=1}^N \alpha_a B_a, \quad \alpha_a \geq 0, \quad \sum_a \alpha_a = 1,$$

where every B_a is effective rational, $K_X + B_a$ is rational Cartier, and (X, B_a) is klt, respectively lc. The decomposition is on the original X .

Proof. We construct a rational affine parameter space that records the boundary and a Cartier presentation simultaneously, and then impose the discrepancy inequalities on one resolution.

Step 1. We construct the finite rational affine parameter space. Write $B = \sum_{i=1}^m b_i S_i$ with distinct prime S_i and $b_i > 0$; the list is empty when $B = 0$. Fix an equality of actual real Weil divisors

$$K_X + B = \sum_{j=1}^p u_j D_j,$$

where the D_j are integral Cartier and a canonical Weil divisor K_X has been chosen. Define V to be the vectors $z = (x_1, \dots, x_m, y_1, \dots, y_p)$ satisfying

$$(12.4) \quad K_X + \sum_i x_i S_i = \sum_j y_j D_j.$$

Coefficient comparison on the finite union of supports gives affine equations with integer coefficients and constants. The space contains $z_0 = (b_i, u_j)$. Gaussian elimination over $\mathbb{Q}$ supplies a rational point and a rational basis of the translation space, so V is rational affine and its rational points are dense. If it has dimension zero, z_0 is rational.

Step 2. We express klt and lc singularities by finitely many rational inequalities. Choose a projective log resolution $\pi: U \rightarrow X$ of X and the finite support in (12.4). Choose K_U compatibly and include all exceptional divisors and the supports of K_U and π^*D_j in one simple normal crossing divisor with components $F_1, \dots, F_s$. For $z \in V$, put

$$(12.5) \quad B(z) = \sum_i x_i S_i, \quad \Gamma_U(z) = \sum_j y_j \pi^* D_j - K_U, \quad K_U + \Gamma_U(z) = \pi^*(K_X + B(z)).$$

Its coefficients $\gamma_l(z)$ are rational affine functions. With $x_i \geq 0$, the resolution criterion [Sat24, Definition 2.1 and Remark 2.3] says klt is equivalent to $\gamma_l(z) < 1$ for every l , and lc is equivalent to $\gamma_l(z) \leq 1$. Its convention is log discrepancy $1 - \gamma_l(z)$; divisors outside the fixed support have discrepancy one. Negative exceptional coefficients are permitted. On a smooth simple normal crossing subpair these inequalities control every further divisorial valuation, so the criterion is a criterion on this given resolution; no separate $\mathbb{Q}$ -Gorenstein hypothesis on X enters.

Step 3. We choose rational vertices without changing the singularity category. For klt (X, B) , the conditions $x_i > 0$, $\gamma_l(z) < 1$ define an open neighborhood of z_0 in V . In rational affine coordinates choose a sufficiently small closed box with rational endpoints containing z_0 and lying in that open. Its finitely many rational vertices express z_0 as a convex combination; in dimension zero use z_0 alone. No x_i conditions occur when $m = 0$.

For lc (X, B) , intersect V with the inequalities $x_i \geq 0$, $\gamma_l(z) \leq 1$, and a bounded closed rational box containing z_0 . This is a bounded rational polyhedron containing z_0 . Its finitely many vertices are rational and their convex hull contains z_0 , also when its dimension is zero. In either case let z_a be the vertices used and $B_a = B(z_a)$. (12.4) expresses $K_X + B_a$ as a rational combination of integral Cartier divisors, while (12.5) ensures the required singularities. Taking the boundary coordinates gives (12.3), and also $K_X + B = \sum \alpha_a (K_X + B_a)$ as real Cartier divisors. $\square$

Proposition 12.3 (Real klt transfer). *Assume CS_n . Let (X, B) be a projective klt real pair of dimension $n \geq 1$, with effective finite-support boundary and real Cartier log canonical divisor. If L is strictly nef rational Cartier and ℓL is integral Cartier, then $K_X + B + tL$ is ample for every real $t > 2n\ell$.*

Proof. Use the klt decomposition of Lemma 12.2. Set $H = \ell L$. Proposition 12.1 gives ampleness of $K_X + B_a + (t/\ell)H$ for $t/\ell > 2n$, for each rational pair on the original X . Their convex combination is $K_X + B + tL$, hence ample. The argument jointly preserves boundary coefficients and a Cartier presentation; it does not approximate them independently. $\square$

Proof of Corollary 1.1. Theorem B gives CS_n in every dimension; for instance $K_V + (2n+1)H$ is ample and hence big on each canonical V . Proposition 12.3 gives the coarse real threshold $2n\ell$. Put $H = \ell L$ and choose $T > 2n$. Then $K_X + B + TH$ is ample. Lemma 2.5, now applied directly to the real klt pair, proves ampleness of $K_X + B + sH$ for every real $s > n+1$. Taking $s = t/\ell$ proves the claimed range $t > (n+1)\ell$. $\square$

13. LOG BIGNESS ON LC STRATA

In this section, we prove Corollary 1.2 by establishing bigness on every lc stratum before applying the Reid–Fukuda basepoint-free theorem.

Definition 13.1. A *quasi-log canonical structure* $[Z, \omega]$ is given by a proper morphism $f: (Y, \Delta_Y) \rightarrow Z$ from a globally embedded simple normal crossing pair, with every coefficient

of Δ_Y at most one, such that $f^*\omega \sim_{\mathbb{R}} K_Y + \Delta_Y$ and the natural map

$$(13.1) \quad \mathcal{O}_Z \xrightarrow{\sim} f_*\mathcal{O}_Y \left(\lceil -\Delta_Y^{\leq 1} \rceil \right)$$

is an isomorphism. Here $\Delta_Y^{\leq 1}$ is the sum of terms of coefficient less than one. Its strata are the images of irreducible components and lc strata of the normalization of this pair. A *rational structure* means the divisors are rational and the equivalence is rational linear equivalence. A divisor on an lc pair is *log big* if its restriction to every lc stratum, including the ambient variety, is big; every divisor on a zero-dimensional stratum is big.

We use [Fuj16, Definitions 3.1 and 3.3] for globally embedded simple normal crossing pairs and quasi-log structures, [Fuj16, Section 2.3] for log bigness, and [Fuj16, Theorem 3.8] for adjunction. This terminology does not assert that an lc center is normal or admits an effective ordinary lc boundary representing the restricted class.

Theorem 13.2 (Quasi-log adjunction, [Fuj16, Theorem 3.8(i)]). *For a projective lc pair (X, B) , $[X, K_X + B]$ is quasi-log canonical, and each lc center Z , with its reduced structure, inherits $[Z, (K_X + B)|_Z]$. The qlc strata in this construction are the lc strata. For rational input the structures are rational.*

The rationality follows by restricting the rational log canonical divisor on the union of resolution strata mapping to Z ; the identification of strata is part of the lc application in [Fuj16, Section 4].

Theorem 13.3 (Normalization, [FL18, Theorem 1.1]). *The normalization $\nu: Z^\nu \rightarrow Z$ of an irreducible qlc pair $[Z, \omega]$ carries $[Z^\nu, \nu^*\omega]$ as a qlc pair. Rational structure is preserved. Its qlc centers map to qlc centers of Z , and its non-quasi-klt locus is the inverse image of that of Z .*

Rational structure is stated in [FL18, arXiv v3, Remark 3.1]; the construction uses Stein factorization of the components dominating Z and restriction of their rational boundaries.

Theorem 13.4 (Normal qlc structure, [Fuj22, Theorem 3.2]). *For a normal qlc variety $[T, \omega]$ there exists a projective birational $p: W \rightarrow T$ with W smooth and*

$$(13.2) \quad p^*\omega = K_W + \Delta_W + M_W,$$

where Δ_W is a simple normal crossing subboundary with exceptional negative part, and M_W is potentially nef. For a rational structure both are rational. If T is projective, W is projective and M_W is nef.

Potentially nef means a finite positive combination of divisors extending to nef divisors on completions, as in [Fuj22, Definition 3.1]. A projective W is already complete, so each such restriction is nef. The negative part of Δ_W need not vanish.

Theorem 13.5 (Reid–Fukuda basepoint-freeness, [Fuj16, Theorem 1.5]). *Let (X, B) be a projective lc real pair and J a nef integral Cartier divisor. If $aJ - (K_X + B)$ is nef and log big for some real $a > 0$, then $\mathcal{O}_X(kJ)$ is globally generated for every sufficiently large integer k .*

Proposition 13.6 (Coarse lc ampleness). *Let (X, B) be a normal integral projective lc real pair of dimension $n \geq 1$, with effective finite-support boundary and real Cartier $K_X + B$. Let L be strictly nef rational Cartier and let ℓL be integral Cartier. Then $K_X + B + tL$ is ample for every real $t > 2n\ell$.*

Proof. We first take rational B , integral Cartier L , and rational $t > 2n$, and put $R = K_X + B$, $D_t = R + tL$. The canonical induction is already complete; no assertion in this proof feeds back into it.

Step 1. We record the available bigness and nefness inputs. Theorem B gives CS_d for every $1 \leq d \leq n$. Thus Theorem 4.3 gives $K_W + sg^*H$ big for every rational $s > 2d$ when W is

smooth of dimension d and g is generically finite onto a positive-dimensional image in a normal projective variety with strictly nef Cartier H .

The lc cone bound in Theorem 2.2, also [Fuj22, Theorem 7.7] for $[X, R]$, gives $R + 2nL$ nef: each negative rational generator has $-R \cdot C \leq 2n$ and $L \cdot C \geq 1$. Hence $D_t = (R + 2nL) + (t - 2n)L$ is nef and strictly nef.

Step 2. We prove bigness on the normalization of every positive-dimensional stratum. Let Z be X or an lc center, with reduced structure, and $d = \dim Z \geq 1$. Its finite normalization $\nu: T \rightarrow Z$ has strictly nef Cartier $H = \nu^*(L|_Z)$: for every integral curve, its degree is the positive degree of the finite map times the original L -degree. Put $\omega_T = \nu^*(R|_Z)$. Theorems 13.2 and 13.3 give a rational qlc structure $[T, \omega_T]$. Theorem 13.4 gives a smooth projective $p: W \rightarrow T$ and rational $\Delta_W = \Delta_W^+ - E$, where $\Delta_W^+, E \geq 0$, E is exceptional, and M_W is nef.

Put $A_T = \omega_T + tH = \nu^*(D_t|_Z)$. Since $t > 2n \geq 2d$, Step 1 gives $K_W + tp^*H$ big, and (13.2) gives

$$(13.3) \quad p^*A_T + E = K_W + tp^*H + M_W + \Delta_W^+ \quad \text{big.}$$

We have kept the nef moduli divisor and the exceptional negative boundary.

Step 3. We descend bigness through the exceptional correction and normalization. For divisible k , $p_*\mathcal{O}_W(kE) = \mathcal{O}_T$: a rational function with only exceptional allowed poles has no divisorial pole on the normal T , and effectivity gives the reverse inclusion. Projection formula identifies

$$(13.4) \quad H^0(W, k(p^*A_T + E)) = H^0(T, kA_T).$$

The left side has positive order- k^d growth by (13.3). Thus A_T is big. It is also nef as the pullback of D_t , so $A_T^d > 0$. The degree-one normalization gives $(D_t|_Z)^d = A_T^d > 0$. The nef top-intersection criterion on the integral projective Z gives bigness of $D_t|_Z$, also when Z is nonnormal; it can be checked after a resolution with birational invariance of bigness. Zero-dimensional strata impose no additional condition. We have proved D_t nef and log big on every lc stratum, not just on X .

Step 4. We apply lc basepoint-freeness with its log-bigness hypothesis. Choose $m > 0$ such that $J = mD_t$ is integral Cartier and put $a = 2/m$. Then

$$(13.5) \quad aJ - R = 2D_t - R = D_t + tL.$$

It is nef, and its restriction to every positive-dimensional lc stratum is big by Step 3 and nefness of tL . Theorem 13.5 makes $|kJ|$ basepoint-free for large k . Its morphism to its reduced projective image is finite: a positive-dimensional projective fiber contains an integral curve, whose J -degree would be zero, contrary to strict nefness. Therefore kJ , J , and D_t are ample.

For real $t > 2n$ with rational B , choose rational $2n < s < t$. Add the nef $(t - s)L$ to the ample $R + sL$ to obtain ampleness of $R + tL$.

Step 5. We pass to real boundaries and the rational polarization. Use the lc case of Lemma 12.2, which keeps the finite Cartier presentation and the closed discrepancy inequalities on the original X . For real $t > 2n$, each rational pair gives an ample $K_X + B_a + tL$. Their finite convex combination is $K_X + B + tL$ and is ample. Finally set $H = \ell L$. The result for integral Cartier H applies at parameter $t/\ell > 2n$, proving the stated coarse bound. No separate rational Cartierness of K_X or $\mathbb{Q}$ -factoriality is introduced. $\square$

Proof of Corollary 1.2. Put $H = \ell L$ and choose $T > 2n$. Proposition 13.6, whose argument proves log bigness on every lc stratum before basepoint-freeness, gives $K_X + B + TH$ ample. Lemma 2.5 uses the sharp lc extremal-ray bound to make $K_X + B + (n + 1)H$ nef. Its positive ample-nef interpolation proves ampleness at every real parameter $s > n + 1$, directly for the real boundary. Taking $s = t/\ell$ proves Corollary 1.2. $\square$

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